

Scale, Yield, and Run Risk in Yield-Bearing Stablecoins

Swaminathan Balasubramaniam*

First prepared: 2025-12-21

This version: 2026-08-20

Abstract

We study the fragility and design of yield-bearing stablecoins: digital dollar claims that combine money-like convenience with returns generated by an underlying trading strategy. These instruments are run-prone because redemptions can impose price pressure and balance-sheet adjustment costs on remaining holders. But they differ from standard fixed-value claims and tokenized deposits because the source of yield may be scale-dependent. We develop a model in which exits affect both redemption pressure and future yield. When yield declines with scale, redemptions can relieve crowding, raise the future value of stayers, and create strategic substitutability; when yield rises with scale, the same feedback amplifies run incentives. Applying the model to Ethena's USDe using daily data from August 2024 to January 2026, we find that supply expansions compress yield, redemptions are sensitive to yield deterioration, and outflows do not predict lower future NAV. We then develop two extensions. A par-preserving protocol can create synthetic substitutability through reward vesting and exit queues, even with scale-neutral yield. Natural scale relief can also create a medium-scale trap: the same decreasing returns that stabilize redemptions can limit adoption by compressing investor yield. Yield-bearing stablecoins should therefore be evaluated by both their yield technology and redemption-reward design.

Keywords: Stablecoins, Yield, Perpetual Futures, Limits to Arbitrage

JEL: G10, G12, G41, C55.

*NEOMA Business School, Rouen, France; s.balasubramaniam@neoma-bs.fr. I thank Gilles Chemla and participants of the 3rd Structured Retail Products and Derivatives Conference 2026 for useful suggestions.

1 Introduction

Stablecoins are increasingly the settlement layer of crypto markets. They provide the stable unit of account that allows blockchains to be used for trading, collateral, settlement, and payments. Yet stablecoins are also inherently fragile. A stablecoin is a promise that a token remains worth one dollar, and that promise can be run-prone: if holders expect others to redeem first, they may rush to exit before the assets that back the claim become illiquid or lose value. This is the familiar logic of runs in fixed-value claims. It is also the reason stablecoins have become a central object of regulatory concern.

This regulatory concern has become sharper as stablecoins compete with tokenized deposits and as some stablecoins move beyond payment claims into yield-bearing structures. Recent regulatory frameworks increasingly draw a line between payment stablecoins, which are generally prohibited from paying interest, and bank deposits, which may pay interest because they remain inside the banking perimeter ([European Parliament and Council of the European Union, 2023](#); [United States Congress, 2025](#)). This raises a policy question that is central to our analysis: is the restriction on stablecoin yield mainly a stability requirement, or does it reflect a broader decision to keep interest-bearing digital money inside the banking perimeter? The answer depends not only on whether a digital dollar pays yield, but on how that yield is produced and how the redemption mechanism is designed. Tokenized deposits pay interest from a bank balance sheet that performs maturity transformation. Yield-bearing stablecoins may instead generate returns from shorter-horizon trading or arbitrage strategies. These different yield technologies can create different forms of fragility, and the redemption design of a protocol can amplify or dampen those fragilities independently of the yield source.

This paper studies the fragility and design of yield-bearing stablecoins. The central point is that redemptions can affect both the liability side of the stablecoin and the profitability of the strategy that supports its yield. If the yield technology is mechanical and scale-sensitive, exits need not only impose liquidation pressure.¹ They can also reduce the size of a crowded strategy and improve the

¹Ethena's USDe is a leading example. In Ethena, issuance and redemptions mechanically change the size of a collateralized spot position and an offsetting short perpetual-futures hedge. When the protocol becomes larger, the hedge must also become larger; when holders exit, the hedge shrinks. Section 2 explains the perpetual-futures mechanics and why this can make the continuation yield scale-sensitive.

continuation yield for those who remain.² When this scale-relief force is strong enough, the yield technology can partly soften fragility by making exits strategically substitutable. Further, we show that even when the yield technology is scale-neutral, a protocol can create an analogous stabilizing force synthetically through reward design.

The paper’s contribution is twofold. First, we show that the yield source can alter the strategic structure of stablecoin redemptions. In standard run models, others’ exits make an individual investor’s own exit more attractive because redemptions impose price pressure and adjustment costs on those who remain (Ma et al., 2025; Gorton et al., 2025). In a yield-bearing stablecoin with decreasing returns to scale, the same exit flow can also raise the continuation yield by shrinking the underlying strategy. This natural scale-relief channel creates strategic substitutability. Second, after testing this channel in the data, we use the logic of the model to derive two extensions. The first is a par-preserving reward-design mechanism. Even when the yield technology is approximately scale-neutral, a protocol can create synthetic substitutability through reward vesting and exit queues: exiting holders redeem at par but forfeit unvested rewards, which accrue to continuing holders. The optimal mechanism chooses the least costly combination of exit delay and unvested reward needed to eliminate local run amplification. The second extension is a scalability implication. The same decreasing-returns technology that stabilizes exits by relieving crowding can limit adoption by compressing the yield that attracts investors. Yield-bearing stablecoins may therefore be robust as medium-scale structured products, but difficult to scale into broad payment instruments when the underlying yield strategy is capacity constrained.

We study the natural scale-relief channel theoretically and empirically using Ethena’s USDe. Ethena’s yield is generated by a delta-neutral strategy that combines spot crypto exposure with short perpetual-futures positions. As USDe supply grows, the protocol must maintain a larger short futures book; as USDe is redeemed, that book shrinks. A seven-day cooldown separates the announcement of exit intent from the execution of redemptions, creating a useful institutional setting for studying the timing of market responses. This setting is well suited to testing the natural

²The assumption that returns decline with capital is standard in models and evidence on capacity-constrained investment strategies. Berk and Green (2004) provide the canonical equilibrium argument that flows compete away performance. Empirically, Chen et al. (2004) show that fund performance declines with size; Pástor et al. (2015) document decreasing returns at the active-management industry level; Ramadorai (2013) provides hedge-fund evidence; Naik et al. (2007) show that capital inflows predict lower subsequent alpha; McLemore (2019) finds that fund mergers lead to performance deterioration; and Forsberg et al. (2022) show that performance is related to cohort size.

scale-relief channel because the yield technology is capacity-constrained.

The empirical analysis supports four predictions of the natural scale-relief model. Supply expansions predict lower future yield, consistent with crowding of the carry trade. Exits are more sensitive to carry deterioration when carry is already weak, consistent with a run threshold that shifts sharply in low-carry states. Realized exits do not predict lower future net asset value for remaining stakers, consistent with scale relief offsetting fire-sale dilution. Finally, the cooldown generates a two-stage market response, with prices and yields adjusting both when exit intent is announced and when redemptions are executed. We also use the passage of the GENIUS Act as an institutional shock associated with a large expansion in USDe supply to examine whether supply growth compresses subsequent yield.

The extensions then broaden the analysis beyond the natural scale-relief channel tested in Ethena. The reward-design extension shows how a protocol can create an analogous stabilizing force when the yield technology is more scalable. This mechanism is related to swing pricing and redemption-fee tools in open-end funds ([Capponi et al., 2020](#); [Jin et al., 2022](#); [Matta et al., 2026](#)), and to exit-queue designs studied in proof-of-stake and blockchain protocols ([Neuder et al., 2024](#)). The difference is that the mechanism preserves par redemption: exiters receive the dollar claim, but give up the unvested reward component of the yield-bearing wrapper. The adoption extension shows the opposite side of natural scale relief. If growth crowds the strategy and compresses yield, the protocol may remain bounded below the scale required for broad payment use.

The policy implication follows. Yield-bearing stablecoins should not be evaluated solely by whether they pay yield. They should be evaluated by the yield source and the redemption and reward design jointly. Tokenized deposits inherit fragility from balance-sheet maturity transformation. Yield-bearing stablecoins inherit fragility from strategy capacity, margin, market liquidity, and reward design. More scalable yield-bearing stablecoins may need synthetic reward and redemption design rather than relying on natural scale relief. The relevant regulatory question is not simply whether digital money pays interest, but what balance sheet or trading technology supports that interest, and what mechanism governs exit.

1.1 Related literature

Our paper contributes to several literatures. The closest are stablecoin fragility and redemption design, the economics of capacity-constrained strategies, and the design of exit and reward mechanisms in financial and crypto protocols.

The closest paper on stablecoin runs is [Ma et al. \(2025\)](#), who study stablecoin runs in a two-layer market structure with secondary-market trading by retail investors and primary-market redemption by specialized arbitrageurs. We take their two-layer structure as a starting point and ask how the security design of a yield-bearing stablecoin changes the payoff from staying versus exiting. Our mechanism is also related to [Gorton et al. \(2025\)](#), who show that redemptions can support the peg of a non-interest-bearing stablecoin by reducing the supply of lendable tokens and raising the external lending rate. In their model the stabilizing channel operates through an external lending market; in ours it is embedded in the protocol’s own yield technology.

Our setting also builds on the classic run literature. In models of bank runs and liquidity transformation, fixed-value claims backed by imperfectly liquid assets generate first-mover incentives and self-fulfilling liquidation dynamics ([Diamond and Dybvig, 1983](#); [Allen and Gale, 1998](#); [Bernardo and Welch, 2004](#); [Goldstein and Pauzner, 2005](#)). This logic has been applied to money market funds and other open-end vehicles, where redeemability at a fixed value can create run incentives ([Kacperczyk and Schnabl, 2013](#); [Sunderam, 2015](#); [Parlatore, 2016](#); [Schmidt et al., 2016](#)). Digital-currency run risk is studied by [Skeie \(2021\)](#), who shows that widely available digital money can make bank withdrawals faster and more destabilizing. A yield-bearing stablecoin inherits this run logic because redemptions can impose price pressure and balance-sheet costs on those who remain. But it also differs from a conventional fixed-NAV claim because the underlying yield technology can become less crowded when investors exit.

Redemption design is central to this paper’s second contribution. The literature on anti-dilution tools in open-end funds shows that modifying the cost of redemption can reduce first-mover advantages. [Jin et al. \(2022\)](#) show that swing pricing in corporate bond funds reduces fragility by passing trading costs to transacting shareholders. [Lewrick and Schanz \(2023\)](#) provide related evidence that swing pricing dampens redemption sensitivity, while [Li et al. \(2021\)](#) show how discretionary gates in money market funds can create incentives for preemptive redemptions.

[Matta et al. \(2026\)](#) analyze contingent fees in a global-games framework and show that fees reduce run incentives through payoff and strategic-uncertainty channels. Our par-preserving mechanism adapts this redemption-design logic to yield-bearing stablecoins but differs in two ways. First, rather than imposing a redemption fee or price haircut, the mechanism preserves par redemption while reallocating unvested rewards from exiting to continuing holders. Second, the mechanism operates jointly through the reward payoff and through queue length, and we derive the optimal combination of these two instruments. Related contemporary work in blockchain mechanism design treats exit queues as protocol design objects: [Neuder et al. \(2024\)](#) study queue design in proof-of-stake validators.

The scale-relief force links the paper to the literature on capacity constraints and crowded trades. [Berk and Green \(2004\)](#) provide the canonical equilibrium logic: performance attracts flows but additional capital competes away returns when investment opportunities are capacity constrained. Empirically, [Chen et al. \(2004\)](#) show that fund performance declines with size, [Pástor et al. \(2015\)](#) document decreasing returns at the industry level, [Ramadorai \(2013\)](#) provides hedge-fund evidence, [Naik et al. \(2007\)](#) show that strategy-level inflows predict lower alpha, [McLemore \(2019\)](#) uses fund mergers as size shocks, and [Forsberg et al. \(2022\)](#) document cohort-size effects. The existing literature studies the inflow side: additional capital lowers future returns. In our setting, redemptions move the protocol in the opposite direction along the same capacity curve. Exits de-crowd the trade and raise the continuation yield for those who remain. The standard run channel is strategic complementarity; the crowded-trade channel creates strategic substitutability. This is the natural channel the paper documents empirically. The optimal mechanism section then asks how to create an analogous stabilizing effect even when the yield technology is scale-neutral.

Our paper also contributes to the broader stablecoin literature on peg stability, reserve design, and adoption ([Lyons and Viswanath-Natraj, 2021](#); [d’Avernas et al., 2022](#); [Lyons and Viswanath-Natraj, 2023](#); [Gorton et al., 2022](#); [Li and Mayer, 2021](#); [Uhlig, 2022](#); [Routledge and Zetlin-Jones, 2022](#); [Aldasoro et al., 2023](#); [Bertsch, 2023](#); [Liu et al., 2023](#); [Adams and Ibert, 2022](#)). Our contribution is to study yield-bearing stablecoins whose fragility can depend both on the scale properties of the yield technology and on how reward and redemption design interact. Ethena provides an empirical setting in which the yield-bearing claim is backed by a mechanically maintained derivatives strategy.

The analysis also relates to the emerging debate over tokenized deposits and interest-bearing digital money. Payment stablecoin regulation increasingly separates non-interest-bearing stablecoins from bank-issued deposit liabilities. Our framework highlights a different economic distinction: the source of yield and the nature of redemption frictions. Tokenized deposits inherit the liquidity mismatch of banking; yield-bearing stablecoins inherit strategy-capacity, market-liquidity, margin, and reward-design risk. More scalable yield-bearing stablecoins may need the synthetic substitutability from reward design rather than the natural substitutability from crowded-trade relief.

Our empirical setting relates to work on perpetual futures and crypto arbitrage. [He et al. \(2024\)](#) derive no-arbitrage bounds for perpetual futures, while [Christin et al. \(2023\)](#) document returns to spot-perpetual basis trades. In Ethena, the short perpetual position is institutionally inelastic: it adjusts through redemption flows and cooldowns rather than purely through discretionary capital, connecting to limits-to-arbitrage and slow-moving-capital models ([Shleifer and Vishny, 1997](#); [Gromb and Vayanos, 2002](#); [Mitchell et al., 2007](#); [Duffie, 2010](#); [Gârleanu and Pedersen, 2011](#); [Gabaix and Koijen, 2021](#)). We also exploit Ethena’s deterministic cooldown to trace market impact around known future redemptions, linking the evidence to research on predictable flows and announcement effects ([Carhart et al., 2002](#); [Lou, 2012](#); [Shleifer, 1986](#); [Harris and Gurel, 1986](#); [Wurgler and Zhuravskaya, 2002](#); [Chan and Lakonishok, 1995](#); [Keim and Madhavan, 1997](#)). The cooldown separates the revelation of exit intent from the balance-sheet adjustment, allowing us to distinguish information effects from the execution of the hedge unwind.

The remainder of this paper is organized as follows. Section 2 provides institutional background on yield-bearing stablecoins and perpetual futures. Section 3 develops the baseline model, characterizes natural scale relief, and derives empirical predictions. Section ?? develops the optimal reward-vesting and exit-queue mechanism. Section 4 presents the empirical analysis. Section 6 concludes. Appendix ?? studies adoption and the medium-scale trap.

2 Institutional Background on Yield-Bearing Stablecoins

Stablecoins are crypto-assets designed to maintain a stable value, typically relative to the U.S. dollar. Their basic economic role is simple. They give users a digital dollar-like instrument that can move on public blockchains continuously, settle quickly across venues, and serve as the main unit of account inside the crypto ecosystem. In practice, most stablecoin activity remains tied to crypto trading, collateral management, and cross-platform settlement rather than everyday retail payments. Even so, the sector has grown rapidly. BIS documents that aggregate stablecoin market capitalization rose from about \$125 billion to around \$255 billion in less than two years, and other official and policy sources place the market near \$300 billion in 2025, still small relative to the broader financial system but large enough to matter for crypto market functioning and, increasingly, for short-term dollar funding markets.

Yield-bearing stablecoins build on this basic architecture by adding a second promise. In addition to price stability, holders expect some return from the assets or trading strategy that backs the token. Economically, one can think of a stablecoin issuer as operating a yield technology: capital enters the protocol, is deployed into some portfolio or trading strategy, and part of the resulting cash flow is passed through to token holders, either directly or through a staked version of the coin. This design is attractive because ordinary fiat-backed stablecoins generally do not share reserve income with users, even though the issuer may earn interest on the underlying reserve assets. By contrast, yield-bearing designs try to make the stablecoin itself competitive with bank deposits, money market funds, or on-chain lending opportunities while preserving the transactional convenience of a blockchain-native dollar instrument. Official U.S. statements in 2025 continued to treat conventional payment stablecoins as products that do not pay interest to holders, underscoring why a separate market segment for yield-bearing alternatives emerged in the first place.

The recent growth of yield-bearing stablecoins reflects both demand and supply forces. On the demand side, crypto investors increasingly want to hold dollars on-chain without giving up all return, especially in an environment where Treasury bills, money market instruments, staking rewards, and basis-trading strategies can generate meaningful yield. On the supply side, stablecoin issuers have become more sophisticated in packaging these income streams into tokenized forms that are easy to hold, transfer, and use as collateral. Some products invest in conventional reserve

assets and pass through part of that income. Others rely on crypto-native strategies such as secured lending, staking, or delta-neutral hedging against perpetual futures. The segment remains much smaller than the full stablecoin universe, but it has become economically significant. For example, public market trackers place Ethena's USDe near \$6 billion in early 2026, making it one of the largest yield-bearing dollar tokens and one of the largest stablecoins overall.

For understanding our paper, the most important distinction is between reserve-backed payment stablecoins and yield-bearing stablecoins whose backing depends on an active balance-sheet strategy. In a conventional payment stablecoin, the key institutional question is whether reserves are safe and redeemable at par. In a yield-bearing stablecoin, an additional question appears: where does the yield come from, and how elastic is that yield when the protocol scales up or down? This matters because the return paid to holders need not be technologically constant. If the underlying strategy becomes crowded, the marginal yield can decline as issuance expands. Likewise, if redemptions force the protocol to adjust positions only gradually, exit may affect both the future yield of remaining holders and the short-run stability of the peg. These issues are central in derivatives-backed structures, where the stablecoin is supported not just by collateral but by a maintained hedge that must be rolled and managed over time.

The regulatory treatment of these products has also begun to diverge across jurisdictions. In the European Union, MiCA creates a formal framework for stablecoins, distinguishing in particular between e-money tokens linked to a single official currency and asset-referenced tokens linked to other references or baskets. Crucially for our setting, MiCA prohibits issuers of e-money tokens, as well as crypto-asset service providers dealing in them, from granting interest to holders. That rule pushes the European framework toward payment stablecoins and away from straightforward yield-sharing designs at the token level. In the United States, by contrast, the policy discussion in 2025 moved toward establishing a federal prudential framework for payment stablecoins, culminating in the GENIUS Act. At the same time, the relevant U.S. official statements continued to define the core payment stablecoin category around instruments that are fully backed, redeemable one for one, and do not pay interest to holders. The practical result is a growing split. Europe has moved earlier and more explicitly to regulate stablecoins within a harmonized framework that is less hospitable to token-level yield. The United States has moved toward a payment-stablecoin framework while leaving more room, outside that category, for experimentation with products whose economic

substance looks more like an investment or structured cash management vehicle than pure digital money.

This background helps motivate the economic question in our paper. Yield-bearing stablecoins are not simply digital dollars with a higher coupon. They are hybrid instruments that combine monetary convenience, asset management, and market-making or hedging activity. Their stability therefore depends not only on redemption rules and reserve quality, but also on the structure of the underlying yield technology and on the institutional frictions that govern how quickly that technology can be resized. In the application we study, those frictions arise from a derivatives hedge and a protocol-imposed cooldown. More broadly, the same logic applies whenever a stablecoin promises both immediate dollar-like usability and returns generated by a balance-sheet strategy whose profitability and adjustability depend on market conditions.

2.1 Perpetual Futures and the Ethena Yield Technology

Ethena’s USDe is the main empirical setting in this paper. USDe is a synthetic dollar claim backed by crypto collateral and hedged with short derivatives positions. Its staked version, sUSDe, accrues protocol rewards through an increasing USDe-per-sUSDe exchange rate. The core yield technology is a delta-neutral spot-perpetual strategy. In simple terms, the protocol holds long exposure to Ethereum and takes an offsetting short position in Ethereum perpetual futures.³ The long position gives the protocol exposure to the crypto asset, while the short perpetual position is designed to hedge the dollar value of that exposure. If the hedge is exact, changes in the Ethereum price largely offset across the long and short legs, leaving the protocol exposed mainly to the net carry from the strategy rather than to the direction of the Ethereum price.

Perpetual futures differ from standard dated futures because they have no maturity date. Instead of converging to spot at expiration, the contract is kept close to the spot price through periodic funding payments between long and short positions. When the perpetual price trades above the spot price, funding is typically positive: long positions pay short positions. When the perpetual price trades below the spot price, funding is negative: short positions pay long positions. A trader who is long Ethereum and short the Ethereum perpetual therefore earns positive funding when leveraged

³In practice, Ethena uses staked Ethereum and related liquid-staking assets on the long side, so the strategy can earn both staking rewards and derivatives carry. The example uses Ethereum spot for simplicity.

demand for long perpetual exposure is strong enough that longs pay shorts. In the simplified example below, we abstract from staking rewards and directional price bets, so the yield of the strategy comes only from the funding rate on the short perpetual position.

A simple numerical example illustrates both the hedge and its scale dependence. Suppose the protocol issues \$100 of USDe, holds \$100 of Ethereum exposure, and shorts \$100 of Ethereum perpetual futures. If the Ethereum price rises by 10 percent, the long Ethereum leg gains about \$10 while the short perpetual leg loses about \$10. If the Ethereum price falls by 10 percent, the long Ethereum leg loses about \$10 while the short perpetual leg gains about \$10. The hedge is therefore designed to make the net position approximately insensitive to the direction of Ethereum prices. The remaining return is the funding payment on the perpetual. For example, if the annualized funding rate paid by longs to shorts is 8 percent, a \$100 short hedge generates roughly \$8 of annualized funding income, before fees, reserves, and other costs.

The same example shows why issuance can mechanically affect continuation yield. If USDe supply doubles from \$100 to \$200, the protocol must support the larger dollar claim with roughly \$200 of Ethereum exposure and a roughly \$200 short perpetual hedge. If the market has abundant demand for levered long perpetual exposure, the short hedge may continue to earn an 8 percent funding rate. But if the required short hedge becomes large relative to long demand, the funding rate can be competed down. For example, the funding rate might fall from 8 percent to 5 percent as the hedge grows from \$100 to \$200. The protocol then earns more dollars of total funding income, but a lower yield per dollar of USDe. Conversely, if \$50 of USDe is redeemed and the hedge shrinks from \$200 to \$150, the reduced short pressure can relieve crowding in the perpetual market and raise the funding rate available on the remaining position.

This is the institutional mechanism behind the scale-relief channel in the model. Exits may impose redemption pressure, but they can also shrink a crowded yield-generating strategy. When the short perpetual-futures book is large relative to demand from longs, redemptions can improve the continuation yield earned by remaining sUSDe holders by reducing the size of the hedge that must be maintained.

3 A Model of Yield-Bearing Stablecoins

The defining feature of a yield-bearing stablecoin is its yield-bearing technology: the protocol generates a return for investors that may depend on the scale of outstanding liabilities. This dependence has direct implications for both the incentive to enter the protocol and the continuation value of existing holders. Unlike traditional fiat-backed stablecoins, in which reserve income is largely scale-neutral, and distinct from tokenized deposits, whose yield is tied to a bank balance sheet, a yield-bearing stablecoin can expose investors to strategic interdependencies in the exit decision that arise through the scale sensitivity of the yield technology. The model developed here makes these forces precise. After developing the main stablecoin mechanism, we use the same primitives to compare yield-bearing stablecoins with tokenized deposits.

The baseline model isolates the natural scale-relief channel: when the yield technology has decreasing returns to scale, redemptions de-crowd the underlying strategy and raise the continuation yield for those who remain. Section ?? then asks how a protocol can synthesize the same stabilizing force when the yield technology is closer to scale-neutral—through reward vesting and exit queues rather than through a naturally capacity-constrained trade.

We follow [Ma et al. \(2025\)](#) (hereafter MZZ) in distinguishing between two categories of agents that interact through a two-layer market structure. Retail investors hold the stablecoin and may sell in the secondary market if they wish to exit. Arbitrageurs, analogous to the authorized participants in MZZ, are the agents with direct access to the protocol’s redemption mechanism. Our setting extends MZZ by adding a yield-generating technology whose returns depend on the scale of the outstanding liability, which introduces an additional channel through which exit decisions affect the continuation value of remaining investors. In MZZ, selling pressure expressed as a fraction of outstanding supply is the natural object, because their mechanism operates through secondary-market price impact and reserve illiquidity. In our setting, however, the absolute scale of the protocol enters the continuation payoff directly through the scale-dependent yield technology, and a larger protocol unwinding a given exit fraction implies a larger redeemed notional and, consequently, greater operational adjustment costs. It is therefore natural to express both the secondary-market discount and the protocol-side adjustment cost as functions of the redeemed notional fQ , where $f \in [0, 1]$ is the exit fraction, Q is outstanding scale, and fQ is the notional volume redeemed. We

retain the MZZ spirit of a two-layer market structure and a reduced-form secondary-market wedge, adapting their argument to a setting in which protocol size matters directly.

We now describe the three primitives of the model in turn.

Yield technology. The protocol generates a one-period yield $y(Q)$, which is continuously differentiable in the scale Q of the outstanding liability. No sign restriction is imposed on $y'(Q)$ at this stage. When $y'(Q) < 0$, larger scale crowds the yield-generating technology and compresses the return available to investors; this is the relevant regime for protocols whose profitability depends on occupying a finite opportunity in derivatives or money markets. When $y'(Q) \approx 0$, the yield is approximately independent of scale, as in reserve-backed designs in which interest accrues on a pool of safe assets irrespective of the number of outstanding tokens. When $y'(Q) > 0$, the yield technology exhibits increasing returns to participation, a case that arises when network effects or broader adoption increase the value generated by the protocol. After exits of fraction f , the redeemed notional is fQ , the remaining scale is $Q(1 - f)$, and the yield available to continuing investors is $y(Q(1 - f))$.

Retail investors. Retail investors hold the stablecoin and cannot redeem directly with the protocol. Their only exit channel is the secondary market, where they sell at a discount $k(fQ)$ below par. The discount is a function of the redeemed notional fQ : the depth of the secondary market is strained by the absolute volume of tokens offered for sale, so a given exit fraction generates greater price pressure the larger the outstanding supply. For a generic notional argument $x \geq 0$, the discount satisfies $k(0) = 0$ and $k'(x) > 0$; the relevant argument here is $x = fQ$.⁴ The payoff of a retail investor who sells at $t = 1$ is

$$W_L(f) = 1 - k(fQ).$$

Arbitrageurs. Arbitrageurs purchase the stablecoin in the secondary market from selling retail investors and redeem it with the protocol at par, receiving one dollar per unit on redeemed notional fQ . Redemption triggers the unwinding of the protocol's underlying positions and generates an operational adjustment cost $C(fQ)$. Expressing adjustment costs as a function of redeemed

⁴The reduced-form discount follows the spirit of MZZ in capturing limited intermediation by arbitrageurs in the secondary market. In a richer formulation one could write the discount as $k(fQ/A)$, where A denotes arbitrage capital or market depth. We suppress A throughout and write $k(fQ)$ for parsimony, treating arbitrage capacity as fixed in the background.

notional is natural because a larger protocol holds larger positions, and unwinding a given exit fraction of a larger book entails greater operational expense and potentially greater market impact. For a generic notional argument $x \geq 0$, the cost function satisfies $C(0) = 0$, $C'(x) > 0$, and $C''(x) \geq 0$; the relevant argument here is $x = fQ$. Arbitrageurs bear a fraction $\tau \in [0, 1]$ of this cost, with the remaining fraction $1 - \tau$ passed on to continuing investors through dilution of the pool. The parameter τ governs the incidence of protocol-side adjustment costs between the arbitrageur redemption channel and the remaining investor base.

The timing of the model is as follows.

$t = 0$. The protocol is entered at scale Q . Investors mint the stablecoin and the protocol deploys the proceeds into the yield-generating technology.

$t = 1$. A fraction $f \in [0, 1]$ of investors face a liquidity shock and must decide whether to sell in the secondary market, receiving $1 - k(fQ)$ per unit, or to remain invested. Simultaneously, arbitrageurs absorb the secondary-market flow by purchasing in the secondary market and redeeming notional fQ with the protocol at par, incurring adjustment cost $C(fQ)$ in the process of unwinding the underlying positions.

$t = 2$. A portion $(1 - \tau)$ of the adjustment cost $C(fQ)$ is passed on to remaining investors through dilution of the continuing pool, which now operates at reduced scale $Q(1 - f)$.

A retail investor who remains invested holds a claim on the post-redemption pool at $t = 2$. This claim is worth par plus the yield generated by the protocol at the reduced scale $Q(1 - f)$, less the per-unit share of adjustment costs allocated to stayers. Because the total cost passed to the continuing pool is $(1 - \tau)C(fQ)$ and the continuing pool has scale $Q(1 - f)$, the per-unit dilution burden is $(1 - \tau)C(fQ)/[(1 - f)Q]$. The payoff from staying is therefore

$$W_S(f) = 1 + y(Q(1 - f)) - \frac{(1 - \tau)C(fQ)}{(1 - f)Q}.$$

Define the incentive to stay as $\Delta(f) = W_S(f) - W_L(f)$. When $\Delta(f) > 0$, remaining in the protocol dominates selling; when $\Delta(f) < 0$, exit is individually rational.

The strategic structure of the exit decision is governed by how $\Delta(f)$ changes with aggregate exit f . Three distinct forces operate, and their relative magnitudes determine whether exit is self-

limiting or self-reinforcing. First, as more investors sell, the redeemed notional fQ rises, depressing the secondary-market execution price through $k(fQ)$ and making further exit less attractive—a stabilizing, self-limiting force. Second, when $y'(Q) < 0$, a larger exit fraction shrinks remaining scale $Q(1 - f)$, relieving crowding in the yield technology and raising the continuation yield for those who stay; when $y'(Q) > 0$, exits erode scale benefits and generate strategic complementarity. Third, the per-unit adjustment cost burden $(1 - \tau)C(fQ)/[(1 - f)Q]$ rises with f , reducing the staying payoff and pushing toward run dynamics. Relative to MZZ, our model adds a direct role for protocol scale in both the yield technology and the notional size of redemptions: a given exit fraction corresponds to a larger redeemed notional the larger the protocol, amplifying quantitatively both the price-pressure and adjustment-cost channels. The following proposition collects these effects.

Proposition 1 (Strategic interaction). *The slope of the incentive function satisfies*

$$\Delta'(f) = Q k'(fQ) - Q y'(Q(1 - f)) - (1 - \tau) \frac{(1 - f)Q C'(fQ) + C(fQ)}{(1 - f)^2 Q}.$$

Strategic complementarity or substitutability in exit decisions is determined by the relative magnitudes of three forces: the secondary-market price-pressure effect, the scale effect of exits on the yield technology, and the dilution of remaining investors through protocol-side adjustment costs.

The decomposition of $\Delta'(f)$ into its three components is most transparent when written with explicit labels:

$$\Delta'(f) = \underbrace{Q k'(fQ)}_{\text{secondary-market price-pressure force}} - \underbrace{Q y'(Q(1 - f))}_{\text{scale effect through the yield technology}} - \underbrace{(1 - \tau) \frac{(1 - f)Q C'(fQ) + C(fQ)}{(1 - f)^2 Q}}_{\text{dilution from protocol-side adjustment costs}}.$$

The first force, $Q k'(fQ) > 0$, is always stabilizing: rising aggregate selling pressure raises the redeemed notional fQ , depresses the secondary-market execution price, and reduces the attractiveness of further exit. The second force reflects the sign of $y'(Q)$: when $y'(Q) < 0$, exits relieve crowding in the yield technology by shrinking remaining scale, raising the continuation yield for those who stay and reinforcing stability; when $y'(Q) > 0$, exits destroy scale benefits, pushing toward strategic complementarity and run dynamics. The third force is always destabilizing: protocol-side adjustment costs borne by remaining investors reduce the continuation payoff

from staying, making exit relatively more attractive as f rises. Both the first and third forces become more quantitatively important as protocol scale rises, because a given exit fraction corresponds to a larger redeemed notional the larger the outstanding liability—a channel absent from models in which costs depend only on the exit fraction f .

Proposition 1 also motivates the design problem studied in Section ???. The natural stabilizing term is $-Q y'(Q(1-f))$, which is positive when $y'(Q) < 0$. The optimal reward-vesting mechanism creates an analogous stabilizing force by reallocating unvested rewards from exiting claims to continuing claims, even in the scale-neutral benchmark $y'(Q) = 0$.

The equilibrium stability of yield-bearing protocols is therefore fundamentally tied to the sign of $y'(Q)$. The potential unraveling of a crowded trade has been emphasized by Stein (2009) in the context of hedge-fund strategies, arguing that capital inflows compress returns and that the resulting fragility can amplify systemic risk when positions are unwound. The same logic applies here at the protocol level: the sensitivity of yield to scale determines whether exit is self-limiting or self-reinforcing. Relative to a pure MZZ-type run model, our framework adds a direct role for protocol scale in the continuation payoff through the yield technology, so that the severity of a potential cascade depends not only on the exit fraction but on the absolute size of the positions being unwound.

The decomposition suggests three useful benchmark regimes, summarized in Table 1. First, *scale-neutral* assets ($y'(Q) \approx 0$), such as USDT and USDC, operate in deep credit markets where the marginal cost of deploying additional reserve capital is negligible, so the reserve income generated by the asset side is largely insensitive to the size of the outstanding supply, even if that income is not passed through to token holders. Second, *scale-relieving* assets ($y'(Q) < 0$), typified by Ethena (*USDe*), maintain yield through strategies that become less crowded as outstanding supply contracts. In these protocols the withdrawal of capital improves the yield for remaining participants, acting as a partial stabilizing force against runs. Third, *network-dependent* assets ($y'(Q) > 0$), including algorithmic and incentive-heavy designs, exhibit strategic complementarity in exit decisions. In these cases a reduction in protocol scale erodes the utility or collateralization efficacy of the system, potentially triggering self-reinforcing liquidity cascades. Such dynamics can arise even in the short run in protocols where returns increase with the size of the investor base, making these designs more prone to self-reinforcing exit dynamics.

Table 1: Stablecoin Yield Regimes and Market Size (March 2026)

Regime	Mechanism	Representative Assets	Mkt Cap (\$B)
$y'(Q) \approx 0$	Scale-neutral	USDT, USDC, PYUSD	260.5
$y'(Q) < 0$	Crowded trade / relief	USDe (Ethena); other capacity-constrained designs	11.4
$y'(Q) > 0$	Network / incentive-driven	Algorithmic or incentive-heavy designs	–

Note: Data reflect estimated circulating supply as of March 2026. The scale-relieving category is illustrative and centered on derivatives-backed and capacity-constrained yield-bearing designs. The network/incentive-driven category is conceptual: it captures designs in which adoption or incentive mechanisms raise the return or utility of participation at larger scale.

3.1 A Unique Equilibrium Exit Threshold

To determine equilibrium exit behavior, we introduce uncertainty in the yield environment. The yield function takes the form $y(Q(1 - f), \theta) = \bar{y}(Q(1 - f)) + \theta$, where θ is a fundamental state that shifts the level of the yield. Retail investors do not observe θ directly; instead, each investor i observes a private signal $s_i = \theta + \varepsilon_i$, where ε_i is independently distributed noise with full support. Investors use this signal to decide whether to sell in the secondary market or remain invested.

This information structure is the global-games environment of [Morris and Shin \(1998\)](#). The relative payoff of selling versus staying, $W_L(f, \theta) - W_S(f, \theta)$, is strictly decreasing in θ for any fixed f : higher fundamentals raise the continuation yield and make staying relatively more attractive. The dominance regions required by the global-games argument are therefore non-empty. For θ sufficiently low, selling dominates staying regardless of beliefs about others; for θ sufficiently high, staying dominates selling. Under small and diffuse signal noise, iterated elimination of dominated strategies selects a unique threshold equilibrium.

Proposition 2 (Threshold equilibrium). *There exists a unique threshold equilibrium characterized by a signal cutoff s^* such that each retail investor sells in the secondary market if and only if $s_i < s^*$. The equilibrium exit fraction is determined by the measure of investors whose signals fall below this threshold.*

The sign of $y'(Q)$ has direct consequences for the location of the threshold. When $y'(Q) < 0$, exits reduce outstanding scale and raise the continuation yield available to remaining investors.

This negative feedback acts as a stabilizing force: an investor who contemplates selling recognizes that those who remain will earn a higher return, reducing the likelihood of further exit. The equilibrium run threshold therefore shifts inward as the sensitivity of yield to scale increases in magnitude. When $y'(Q) > 0$, the feedback runs in the opposite direction, potentially generating self-reinforcing selling pressure.

Proposition 3 (Scale relief and stability). *Suppose that $y'(Q) < 0$. Then an increase in the sensitivity of yield to scale reductions lowers the run threshold:*

$$\frac{\partial \theta^*}{\partial (-y'(Q))} < 0.$$

Stronger yield relief following reductions in scale stabilizes the protocol by raising the continuation value of staying and reducing retail investors' incentives to sell in the secondary market. The mechanism-design section shows that an analogous inward shift in the run threshold can be achieved by reward vesting even in the scale-neutral benchmark $y'(Q) = 0$.

3.2 Comparison with Tokenized Deposits

The same primitives also clarify how yield-bearing stablecoins differ from tokenized deposits. A tokenized deposit is an interest-bearing digital dollar claim, but its yield is generated by a bank balance sheet rather than by a protocol-level trading strategy. In the notation of the model, let $y_D(Q)$ denote the deposit yield. A simple benchmark is

$$y_D(Q) \approx i_D, \quad y'_D(Q) \approx 0,$$

where i_D is the deposit rate. The key source of fragility is then not scale relief in the continuation yield, but the balance-sheet cost of meeting withdrawals.

Let $C_D(fQ)$ denote the cost incurred when withdrawals of notional fQ force the bank to use liquid buffers, raise funding, sell securities, or adjust longer-term assets. We assume $C'_D(fQ) > 0$ and $C''_D(fQ) \geq 0$. The incidence parameter $\tau_D \in [0, 1]$ captures the extent to which bank capital, deposit insurance, liquidity regulation, or lender-of-last-resort support absorbs these costs before they are passed to remaining claimants; a higher τ_D corresponds to a more generous backstop. A

tokenized-deposit analogue of the staying payoff is therefore

$$W_{D,S}(f) = 1 + y_D(Q(1 - f)) - \frac{(1 - \tau_D) C_D(fQ)}{(1 - f)Q}.$$

The argument $Q(1 - f)$ is retained only for comparability with the stablecoin payoff; in the benchmark $y'_D(Q) \approx 0$, withdrawals do not materially raise the continuation deposit yield. If the tokenized deposit is directly redeemable at par with the issuing bank, the exit payoff is $W_{D,L}(f) = 1$. This normalization treats y_D as the continuation return earned by claims that remain in the bank balance sheet; any accrued interest paid equally to exiting and staying depositors can be netted out without changing the exit incentive. If the claim also trades in secondary markets, or if redemption is delayed, restricted, or uncertain, then the exit payoff becomes $W_{D,L}(f) = 1 - k_D(fQ)$, paralleling the stablecoin formulation. The secondary-market discount $k_D(fQ)$ is thus not necessarily central to the tokenized-deposit case: when direct at-par redemption is available, the main exit externality flows through $C_D(fQ)$ rather than through a price wedge. In that direct-redemption benchmark, the stabilizing secondary-market force $Qk'_D(fQ)$ is absent, so private run incentives are disciplined mainly by the incidence parameter τ_D , by bank liquidity buffers, and by public or private backstops rather than by a market price discount faced by exiting depositors.

A further distinction is that the redemption rule for a tokenized deposit need not be programmable in the same way as the redemption rule of a protocol-native stablecoin. The token may circulate on-chain, but redemption remains a claim on the issuing bank and may be governed by banking law, deposit contracts, payment-system rules, and supervisory constraints. We therefore do not assume that tokenized deposits can implement a protocol-level cooldown. If such a mechanism were feasible and legally enforceable, it could be represented in the same reduced-form way as our stablecoin cooldown, with withdrawals processed over T_D periods and adjustment costs $T_D C_D(fQ/T_D)$. Economically, however, such a rule would be closer to a withdrawal delay, gate, or contingent redemption arrangement than to a purely on-chain protocol feature. It may reduce balance-sheet adjustment costs, but it would also trade off against the money-like liquidity that makes deposits valuable and may require explicit contractual or regulatory authorization rather than merely smart-contract implementation.

The incentive to stay is defined analogously as $\Delta_D(f) = W_{D,S}(f) - W_{D,L}(f)$. In the direct-

redemption benchmark with $k_D(fQ) \approx 0$ and $y'_D(Q) \approx 0$, the exit externality is governed primarily by $C_D(fQ)$ and τ_D . Withdrawals activate balance-sheet adjustment costs—drawing on liquidity buffers, refinancing, or asset sales—but do not mechanically de-crowd a finite trading strategy and raise the yield to remaining depositors. This is the key structural difference:

Tokenized deposit: withdrawals activate balance-sheet adjustment costs.

Yield-bearing stablecoin: withdrawals may also change the continuation yield through

$$y(Q(1 - f)).$$

It follows that the two claims have different fragility profiles. For a tokenized deposit backed by a conventional bank balance sheet, fragility arises primarily from maturity transformation: the inability to call long-duration assets immediately at par means that large withdrawals force costly balance-sheet adjustment, and the severity of the crisis depends on the depth of the liquid buffer and the generosity of the backstop τ_D . For a yield-bearing stablecoin, fragility additionally depends on the sign and magnitude of $y'(Q)$: when the strategy is capacity-constrained and $y'(Q) < 0$, the same redemptions that impose adjustment costs on stayers also partially relieve crowding and improve the continuation yield—a stabilizing channel that is not present in the simple tokenized-deposit benchmark unless the bank’s asset returns themselves improve when the deposit base shrinks.

Table 2 summarizes the mapping of the three main types of digital dollar claims into the model primitives.

Table 2: Mapping Digital Dollar Claims into the Model Primitives

Claim		Yield source	Key model objects	Main fragility
Payment stable-coin		Liquid reserves	$y'(Q) \approx 0$	Reserve liquidity and confidence
Tokenized deposit	de-	Bank balance sheet	$C_D(fQ)$, τ_D , $k_D(fQ)$	Maturity transformation and backstop design
Yield-bearing stablecoin		Trading/liquidity strategy	$y'(Q)$, $C(fQ)$, $k(fQ)$	Strategy capacity and redemption design

Note: For tokenized deposits, $k_D(fQ)$ is relevant only when the claim trades in secondary markets or when redemption is delayed, restricted, or uncertain. If tokenized deposits redeem directly at par with the issuing bank, the main fragility is instead captured by the balance-sheet adjustment cost $C_D(fQ)$, moderated by the incidence or backstop parameter τ_D .

The remainder of the model focuses on the yield-bearing stablecoin case, where the scale sensitivity of $y(Q)$ is central. The optimal reward-vesting mechanism in Section ?? is most

naturally suited to protocol-native yield-bearing wrappers, where eligibility, vesting schedules, and exit queues can be encoded at the contract level. Tokenized deposits may not be able to implement such mechanisms without explicit contractual or regulatory authorization.

3.3 Derivative-Backed Stablecoins

Ethena provides a setting where the natural scale-relief channel is likely to operate because the yield technology is a capacity-constrained derivatives trade. In the Ethena protocol, yield is generated by a delta-neutral carry strategy and accrues exclusively to staked token holders. The staked token, $sUSDe$, earns yield, while the unstaked token, $USDe$, does not bear yield but can be freely traded on secondary markets. An investor who wishes to exit must first unstake from $sUSDe$ to $USDe$ before selling. The economics of Ethena's carry strategy place it in the scale-relieving regime, $y'(Q) < 0$: larger outstanding supply crowds the carry opportunity and compresses the return available to stakers.

Yield technology. Ethena earns yield by maintaining a delta-neutral position that combines a long holding of spot ETH with a short position in the corresponding perpetual futures contract. The yield arises primarily from the funding rate paid by long-position holders to short-position holders when the market is in contango. As the outstanding supply Q expands, the protocol must maintain a larger aggregate short position in perpetual futures markets. In the empirically relevant range, this larger scale crowds the carry opportunity: the aggregate short position competes for a finite pool of levered long demand, compressing funding rates and reducing the return available to stakers. Consequently, Ethena operates in the regime of decreasing returns to scale, $y'(Q) < 0$. We capture this with the linear specification

$$y(Q) = \bar{y} - \phi Q, \quad \phi > 0,$$

so that $y'(Q) = -\phi < 0$. Redemptions that reduce outstanding supply relieve this crowding and restore yield, consistent with Proposition 3 and Corollary 1.

Corollary 1 (Entry crowding). *Under the specification $y(Q) = \bar{y} - \phi Q$ with $\phi > 0$, an increase in protocol scale reduces equilibrium yield. Consequently, increases in outstanding supply predict deterioration in the perpetual futures basis and lower future yields, while reductions in supply*

through redemption are associated with basis recovery.

Periods of rapid minting, during which Q expands, should therefore be followed by compression of the funding rate basis, while episodes of net redemption should coincide with or be followed by basis recovery as the aggregate short position unwinds.

Retail investors. Holders of $USDe$ who are not whitelisted cannot access the protocol’s mint-and-redeem mechanism directly. Their only exit option is to sell $USDe$ on secondary markets such as decentralized exchanges, where they receive a price below par, $1 - k(fQ)$, when selling pressure—measured by redeemed notional fQ —is elevated.

Arbitrageurs. Whitelisted participants can redeem $USDe$ directly with Ethena at par. The adjustment cost $C(fQ)$ captures the market impact and operational cost incurred when Ethena closes the delta-neutral hedge positions corresponding to the redeemed notional fQ . Because these positions are held in perpetual futures markets, rapid or large-scale unwinding generates price impact that is increasing and convex in the redeemed notional, consistent with the assumed properties of C . Arbitrageurs bear a fraction τ of this cost directly, with the remainder passed to the continuing pool.

Cooldown mechanism. Yield in the Ethena protocol accrues exclusively to holders of staked $sUSDe$. An investor who wishes to exit must first unstake from $sUSDe$ to $USDe$, which triggers a mandatory cooldown period of seven days before the unstaking is completed and the resulting $USDe$ can be transferred or sold. We model this institutional feature in reduced form by assuming T cooldown subperiods, over which total redeemed notional fQ is processed by arbitrageurs in equal tranches of size fQ/T . This design separates the announcement of exit intent from the execution of redemptions and smooths the adjustment costs incurred at each step. Because C is strictly convex, processing a smaller tranche at each date is less costly per unit than processing the full notional at once.

Proposition 4 (Cooldown smoothing). *If C is strictly convex, $C(0) = 0$, and $T > 1$, then spreading total redeemed notional fQ uniformly over T periods in tranches of size fQ/T reduces total protocol-side adjustment costs:*

$$T C\left(\frac{fQ}{T}\right) < C(fQ).$$

Cooldown mechanisms therefore reduce the aggregate cost of processing redemptions by smoothing

the pace at which the protocol unwinds its underlying positions.

The separation of announcement from execution also generates a two-stage pattern in observable market variables. At the time retail investors submit unstake requests, the aggregate intention to exit becomes observable to all market participants. This announcement is informative about the forthcoming volume of arbitrageur redemptions, the expected path of protocol scale, and the trajectory of the funding rate basis. Secondary-market prices and the discount $k(fQ)$ may therefore adjust at the announcement stage before any redemptions are executed. A second and distinct adjustment then occurs as arbitrageurs process each tranche of notional fQ/T and the protocol unwinds the corresponding hedge positions.

Proposition 5 (Two-stage market response). *When exits are implemented through a cooldown mechanism, market variables may respond at two distinct times: first, when retail investors' exit intentions become observable through unstake requests, and second, when arbitrageurs execute redemptions and the protocol unwinds its underlying positions in tranches of notional fQ/T . These two stages generate predictable dynamics in market quantities related to exit behavior.*

The cooldown in Ethena is an observed institutional feature with a fixed seven-day horizon. The mechanism-design section generalizes this by treating the queue length T as a design variable and pairing it with a per-unit unvested reward r , choosing jointly the least costly combination needed to eliminate local run amplification.

3.4 Empirical Predictions

The model yields a small set of sharp empirical predictions that map directly to observable quantities in derivative-backed stablecoins.

Prediction 1: Scale and future yield. In the regime in which the yield technology exhibits decreasing returns to scale, $y'(Q) < 0$, higher outstanding supply should predict lower subsequent yield. When protocol scale expands, the underlying carry trade becomes more crowded, compressing the return available to the remaining pool. Conversely, redemptions that reduce scale—and hence redeemed notional—should be followed by yield recovery. This prediction is the empirical counterpart of Proposition 3 and, in the linear specification, of Corollary 1.

Prediction 2: Exit responds to carry, with stronger sensitivity in low-carry states. Retail exits should be increasing in deterioration of the protocol’s yield environment. In the Ethena application, lower perpetual-futures basis or lower funding rates should predict more cooldown initiations, unstakes, and redemptions. Moreover, this sensitivity should be strongest when carry is already compressed, because in that region the marginal benefit of remaining invested is small and modest further declines in yield are more likely to push investors across the exit threshold s^* . The model therefore predicts state-dependent exit elasticities, with the response of outflows to basis deterioration amplified in low-carry regimes.

Prediction 3: Weak fire-sale pass-through to remaining holders when scale relief dominates. If the stabilizing scale effect from $y'(Q) < 0$ is quantitatively stronger than the destabilizing dilution effect generated by adjustment costs $C(fQ)$, then realized exit flows need not depress the continuation value of remaining investors. In that case, higher burns or unstakes should have little negative effect on subsequent net-asset-value dynamics and may even be followed by higher future yield as the protocol’s aggregate position shrinks and carry compression eases. Put differently, observing more exit today need not imply a lower payoff for those who remain when the dominant channel is relief of crowding in the yield technology rather than fire-sale dilution. This is the key empirical implication of the sign-and-magnitude decomposition in Proposition 1.

Prediction 4: The cooldown generates delayed flow adjustment but attenuates destabilization. When exits are implemented through a mandatory cooldown, market responses should display a two-stage pattern. First, variables tied to investor intent should react when cooldown requests are initiated, because these requests reveal information about forthcoming redemptions, the associated redeemed notional, and future protocol scale. Second, realized redemption flows—processed in tranches of notional fQ/T —should occur with a deterministic lag and should exert a smoother effect on prices and yields than under immediate withdrawal. Accordingly, cooldown initiations should forecast future unstake completions at the protocol-imposed horizon, while the realized redemptions themselves should exhibit weaker price impact than a frictionless exit technology would imply. This follows from the convexity logic in Proposition 4 together with the informational structure of Proposition 5.

Before turning to the data, the next section develops the optimal reward-vesting and exit-queue mechanism as a general par-preserving design for yield-bearing stablecoins. The empirical section

then tests the natural scale-relief predictions in the Ethena setting. Appendix ?? studies an adoption extension and shows that a decreasing-return yield technology can generate a medium-scale trap: the same mechanism that stabilizes exits also limits the equilibrium scale that the protocol can support.

4 Data and Empirical Evidence

4.1 Data and Construction of Variables

We construct a daily panel spanning August 1, 2024 through January 31, 2026, covering 549 calendar days that encompass the full growth and partial retraction of Ethena’s USDe stablecoin. The sample begins at the first date for which a complete panel of on-chain and off-chain variables is available.

We organize the data into four categories. The first covers the *yield technology*. The net asset value (NAV) of sUSDe—the staked wrapper that distributes protocol yield—is reconstructed from `RewardsReceived` events emitted by the `StakedUSDeV2` contract on Ethereum mainnet, retrieved via Dune Analytics. The contract distributes rewards in tranches every eight hours, so we aggregate all events within a UTC calendar day to form the daily NAV series. The daily log NAV return is $r_t = \log(\text{NAV}_t / \text{NAV}_{t-1})$. Because the 8-hour vesting schedule interacts with the UTC day boundary, individual daily returns are noisy: annualized daily returns have a standard deviation of 71% and 13.5% of observations exceed $|50\%|$ annualized. These are purely mechanical day-boundary artifacts. To recover clean variation in the yield technology we use a 30-day rolling average of daily log returns, which reduces the annualized standard deviation to 29% while preserving low-frequency yield dynamics. We also report results using the log NAV level, which is free of day-boundary noise by construction.

The second category covers *protocol flows*. Redemptions take two forms. Direct burns occur when USDe holders redeem through the minting contract. Unstakes occur when sUSDe stakers complete a two-step exit: they first call `cooldownAssets()`, which transfers sUSDe to the `USDeSilo` contract and initiates a mandatory seven-day waiting period, and then call `unstake()` to receive USDe. We identify cooldown-start events as `Withdraw` events on the sUSDe contract where the receiver equals the silo address, and unstake completions as `Transfer` events on the USDe contract where the sender is the silo. The seven-day cooldown was fixed on November 20, 2023 and did not change during our sample. This deterministic lag is central to the identification strategy developed in Section 4.4.

The third category covers *market conditions*. Funding rates and ETH perpetual mark prices come from Binance’s public API; ETH spot prices are from Binance spot markets; open interest

data come from Bybit. The mark-minus-spot basis is $(F_t - S_t)/S_t \times 10,000$ in basis points, where F_t is the ETH perpetual mark price and S_t is ETH spot. The USDe secondary-market price is the volume-weighted average of daily swaps in the Curve Finance USDe-USDC pool. USDe outstanding supply is sourced from the on-chain token contract and is available from November 17, 2024 onward (443 observations).

The fourth category captures collateral-composition variation through three subperiods. Phase 1 (August 1 – December 31, 2024) is the period when ETH and BTC delta-neutral positions dominated the collateral pool. Phase 2 (January 1 – June 30, 2025) is a transition period of mixed collateral. Phase 3 (July 1, 2025 – end of sample) is the period when stablecoin collateral exceeded 50% of the pool, a shift partly triggered by the passage of the GENIUS Act, which we exploit as a natural experiment in Section 4.3. Table A1 presents descriptive statistics.

4.2 Scale and Yield

The model establishes that the sign of $y'(Q)$ is the pivotal design parameter for run stability. A yield technology with $y'(Q) < 0$ produces strategic substitutability in exit decisions: when investors leave, the protocol’s aggregate short position shrinks, carry compression eases, and the yield available to remaining stakers rises. We now test whether USDe’s yield technology satisfies this condition empirically (Prediction 1).

A naive regression of yield on contemporaneous supply is uninformative because both variables respond to the same underlying funding-rate environment: when the perp basis is wide, yields are high and investors mint more USDe, generating a positive correlation that masks any crowding effect. We address this using two complementary approaches. The first uses lagged supply as a predetermined regressor and establishes the sign of the relationship across smoothing horizons. The second exploits the passage of the GENIUS Act on July 18, 2025 as an instrument for supply level, as detailed in Section 4.3.

Table A2 reports OLS regressions of annualized yield on log lagged supply (Panel A) alongside a non-parametric quintile decomposition (Panel B). In the preferred OLS specification in column (3), which uses the 30-day rolling yield and controls for collateral-phase dummies and lagged basis with Newey–West standard errors, the point estimate is $\hat{\beta} = -18.6$ ($t = -2.70$): a doubling of

outstanding supply reduces the 30-day annualized yield by approximately 18.6 percentage points. Column (4) uses the log NAV level as the dependent variable and recovers $\hat{\beta} = -0.030$ ($t = -2.97$), confirming the relationship in cumulative terms. Column (1), using the raw daily log return annualized, yields a positive coefficient: this sign reversal is driven entirely by the day-boundary artifact described above, as confirmed by the progression toward zero in column (2) and the reversal in column (3). Panel B provides non-parametric corroboration. Sorting the supply-valid subsample into quintiles of lagged supply, mean 30-day yield falls monotonically from 15.7% annualized in the lowest quintile (\$1.6 billion mean supply) to -3.5% in the highest (\$9.2 billion). The Spearman rank correlation between lagged supply and 30-day yield is -0.111 ($p = 0.021$) for the full sample, strengthening to -0.38 ($p < 0.001$) within Phase 2 and -0.26 ($p < 0.001$) within Phase 3, where supply variation is greatest. The association is absent in Phase 1, consistent with a growth-phase confound: during the protocol’s initial scale-up, yield and supply both rise together as the yield technology proves itself; once the protocol reaches maturity the crowding mechanism dominates.

4.3 The GENIUS Act as a Natural Experiment

The Guiding and Establishing National Innovation for U.S. Stablecoins Act (GENIUS Act) was passed by the Senate on June 17, 2025 with a bipartisan vote of 68–30, by the House on July 17, 2025 with 308–122, and signed into law on July 18, 2025. The Act established the first federal licensing framework for payment stablecoins in the United States, excluded compliant stablecoins from SEC and CFTC jurisdiction, and created a clear pathway for institutional allocation at scale. For Ethena specifically, it resolved the regulatory uncertainty that had constrained institutional participation and triggered near-immediate large-scale entry.

The causal mechanism is close to mechanical. In the 90 days prior to the House vote, USDe supply averaged \$2.2 billion with daily net issuance of approximately -\$14 million. On the day of the House vote, net issuance was +\$267 million. Over the 77 days following signing, average daily net issuance was +\$122 million, and supply reached a peak of \$11.6 billion on October 3, 2025—a 447% increase from the \$2.1 billion level on July 16. Critically, the 30-day trailing yield on July 14 stood at +18.6% annualized, the highest reading in Phase 3. Capital had been waiting on legal clarity, not yield improvement, confirming that the supply surge is orthogonal to the yield

environment at the time of the shock.

Figure 1 illustrates the event. Yield was +18.6% immediately before signing, turned negative on the signing date itself, and reached a nadir of −49% on August 15, coinciding with the fastest inflow velocity. As inflows decelerated and supply plateaued near \$11.6 billion, yield partially recovered to +5.6% at the October peak. When supply retreated to \$3–4 billion over October and November, yield recovered to +30–40% annualized. The yield curve traces a complete parabola around the supply surge, symmetric on entry and exit—the signature of a technology with decreasing returns to scale.

We formalize this with a two-stage least squares design. The instrument is $Z_t = \mathbf{1}[t \geq \text{July 18, 2025}]$. Table A3 reports the first and second stages together. The first-stage coefficient on Z_t is $\hat{\gamma} = 0.954$ ($t = 8.36$, $F = 69.9$), confirming the surge is strongly predicted by the signing date. The second-stage estimate is $\hat{\beta}_{IV} = -12.0$ ($t = -1.80$, $p = 0.073$). The IV estimate is 37% smaller in absolute value than the OLS estimate of −19.0, consistent with mild upward OLS bias: across the full sample, supply growth tends to coincide with high-yield periods as investors respond to favorable carry conditions, biasing OLS away from zero. At the IV point estimate, a doubling of outstanding supply reduces annualized yield by approximately 12 percentage points, and each additional \$1 billion of AUM reduces annualized yield by roughly 1.1 percentage points at the sample mean.

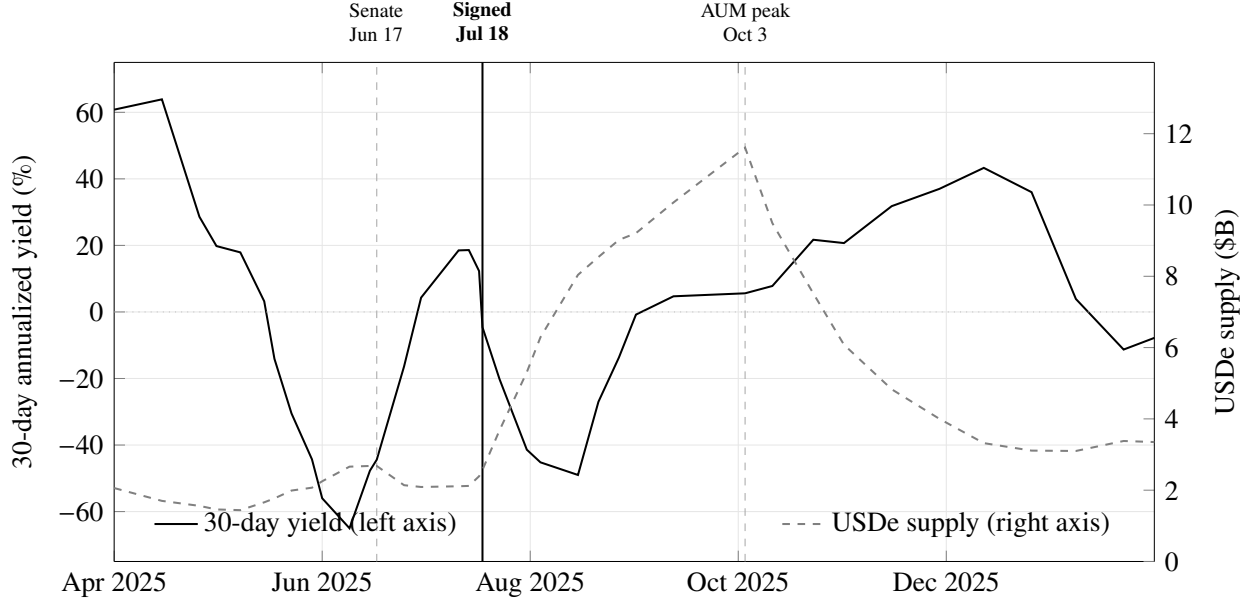
We verify the exclusion restriction with three checks. First, the perpetual basis was −3.2 basis points on July 14 and −3.4 basis points one week after signing—no discrete jump, so the Act did not directly shift funding markets. Second, ETH spot prices were approximately flat over July 14–18, ruling out a concurrent crypto rally. Third, the Phase 3 dummy loads near zero in the first stage ($\hat{\gamma}_{P3} = -0.024$, $t = -0.42$), confirming the instrument does not proxy for the contemporaneous collateral-composition shift. The Act affected yield exclusively through the supply channel.

4.4 Identification: The Cooldown Instrument

Having established that the yield technology exhibits decreasing returns to scale, we turn to whether investor exits generate destabilizing feedback through the fire-sale channel (Prediction 3). The central prediction is that redemptions depress the NAV available to remaining stakers, creating

Figure 1: USDe SUPPLY AND 30-DAY YIELD AROUND THE GENIUS ACT

The solid line is the 30-day rolling annualized sUSDe yield (left axis). The dashed line is outstanding USDe supply in billions (right axis). Vertical lines mark Senate passage (June 17), signing (July 18, bold), and the supply peak of \$11.6B (October 3).



strategic complementarity. The necessary condition is that closing the protocol's short positions imposes costs primarily on those who stay: low adjustment-cost sharing τ in the model's notation.

We test this using the seven-day cooldown as an instrument for unstake completions. Prediction 4 requires that cooldown initiations forecast unstake completions at the protocol-imposed horizon; we verify this condition as a first stage before testing the fire-sale channel. Because investors must call `cooldownAssets()` exactly seven days before redemption becomes eligible, the volume of unstakes completing on day t is determined almost entirely by cooldown initiations on day $t - 7$. Table A4, Panel A reports the first stage. The coefficient on lagged cooldown initiations is 0.869 ($t = 20.82$) in the bivariate specification, stable across controls, with a first-stage F -statistic of approximately 9,461. The same-day correlation between cooldown initiations and unstake completions is only 0.069, confirming the instrument operates through the predetermined lag rather than a spurious contemporaneous association. With a first-stage F -statistic of this magnitude, attenuation bias cannot account for any null results that follow.

4.5 Exit Flows, NAV Dynamics, and Regime Dependence

We test the fire-sale prediction (Prediction 3) by estimating

$$r_{t+k} = \alpha + \beta \text{Flow}_t + \gamma' X_t + \varepsilon_{t+k}, \quad (1)$$

where r_{t+k} is the log NAV return at horizon $k \in \{1, 3, 5, 7, 9\}$ days, Flow_t is either the burn rate or the unstake rate on day t , and X_t includes phase dummies, lagged NAV return, and lagged basis. The fire-sale mechanism requires $\hat{\beta} < 0$: exits today should predict lower future NAV as the protocol unwinds positions at a discount.

Table A4, Panels B and C report OLS and IV results. Across all horizons and both flow measures the OLS coefficient is indistinguishable from zero or positive. At $k = 9$, coefficients are positive and significant, but this reflects the 8-hour reward-vesting artifact: the schedule creates mild positive autocorrelation at multiples of the vesting cycle, unrelated to exit flows. Instrumenting unstakes with cooldown initiations seven days prior (Panel C), IV coefficients are positive at all horizons. The significant positive coefficient at $k = 5$ ($\hat{\beta} = 0.007$, $t = 2.63$) implies that exogenously higher redemptions are associated with *higher* subsequent NAV returns—the opposite of the fire-sale prediction. This result rules out attenuation bias as an explanation for the OLS null. Consistent with Prediction 3, exits reduce protocol scale rather than impose costs on remaining stakers, and the relief effect on crowding dominates any adjustment-cost dilution.

Table A5 examines heterogeneity across collateral phases. Panel A uses burn rates and Panel B uses unstake rates as the exit-flow measure. Exit-flow coefficients are uniformly zero in Phase 1, the ETH and BTC dominant period when position-closure costs are lowest and the fire-sale mechanism should be strongest. The null result in the theoretically cleanest phase is the most direct refutation available. Occasional positive coefficients in Phase 3 are attributable to yield-environment correlation: large redemption events in the stablecoin-dominant phase coincide with high-carry periods when the protocol is growing.

Table A5, Panel C reverses the regression to evaluate Prediction 2. The dependent variable is the cooldown initiation rate at horizon k and the regressor is the current perpetual basis. A lower basis today—less carry available to the protocol—significantly predicts higher cooldown initiations at horizons of 3–9 days, peaking at $k = 7$ ($\hat{\beta} = -0.00045$, $t = -3.05$). Investors

respond to yield signals through a deliberate observation-response mechanism: they observe the basis, assess the forward yield environment, and initiate cooldowns accordingly. This is consistent with Prediction 2’s individual optimization against a market-wide observable and is distinct from strategic complementarity, in which investors run to beat one another.

Finally, Table A6 examines state dependence in exit behavior, directly testing the amplification mechanism in Prediction 2. Column (1) shows that daily burns rise when the basis declines. Column (2) introduces a below-zero basis indicator and its interaction, revealing that exit sensitivity is approximately 11.4 times larger when the basis is below zero ($\hat{\beta}_{\text{interaction}} = -9.42$, $t = -2.49$). This amplification in the low-carry state matches Prediction 2’s implication that marginal investors are most sensitive to further yield deterioration when carry is already compressed. Column (3) shows the mirror pattern at the entry margin: net issuance rises with the funding rate, but the asymmetry for below-zero funding is statistically indistinguishable from zero, indicating regime dependence is more sharply defined on the exit side. Together, exit sensitivity is amplified in the low-carry state, but this operates through individual optimization rather than strategic coordination—the absence of fire-sale dynamics in Table A4 holds throughout the sample regardless of basis regime.

Taken together, the evidence across all six tables points to a single economic narrative. The USDe yield technology exhibits decreasing returns to scale, with $y'(Q) < 0$ of roughly -12 percentage points per log-unit of supply under the GENIUS Act IV. This reflects the protocol’s aggregate short position in perpetual futures compressing the available carry as scale grows. Yet the same negative derivative provides the stabilization property of Proposition 3: exits reduce scale, which raises the yield available to remaining stakers, working against strategic complementarity rather than amplifying it. The data produce uniformly non-negative coefficients in the fire-sale test (Table A4), a first-stage F -statistic of 9,461 that rules out attenuation bias, and a reverse-causality result locating investor exit decisions in deliberate signal-reading rather than panic. Exit sensitivity is amplified in the low-carry state (Table A6), but this amplification operates through individual optimization, not the coordination channel the fire-sale mechanism requires. The cooldown reinforces this stability: by spreading redemptions over seven days, it reduces total adjustment costs through the convexity argument of Proposition 4, further attenuating any destabilizing feedback.

5 Extensions: Design and Scale

The baseline model and the Ethena evidence focus on natural scale relief: when the underlying yield technology has decreasing returns, redemptions can shrink the strategy, relieve crowding, and raise the continuation value of investors who remain. This section develops two implications of that logic. First, a protocol can create a similar stabilizing force through reward and exit design even when the yield technology is approximately scale-neutral. Second, natural scale relief has a scalability cost: the same decreasing-return technology that can stabilize exits can also limit ex ante adoption.

5.1 Reward Vesting and Exit Queues

Consider first a scale-neutral benchmark, $y(Q) = \bar{y}$. In this case, exits do not mechanically raise the underlying yield. A stablecoin protocol that wants to preserve par redemption may also be reluctant to use the standard anti-dilution tools of open-end funds, such as swing pricing or redemption fees, because those instruments reduce the payoff of exiting investors below par (Capponi et al., 2020; Jin et al., 2022). A par-preserving alternative is to separate the base dollar claim from the reward-bearing claim. Investors who enter the exit queue redeem the base stablecoin at par, but forfeit an unvested component of the reward. The forfeited rewards remain in the protocol and accrue to continuing holders.

Fix a stress state with outstanding scale Q and exit fraction $f \in (0, 1)$. The protocol chooses two instruments. The first is an exit-processing time $T \geq 1$. If the exit notional fQ is processed over T equal tranches, the total adjustment cost is

$$TC\left(\frac{fQ}{T}\right),$$

where $C(0) = 0$, $C' > 0$, and $C'' > 0$. The second is a per-unit unvested reward $r \in [0, \bar{y}]$. This reward vests only for holders who remain outside the exit queue.

Under a design (T, r) , the payoff from staying rather than exiting is

$$\Delta^{T,r}(f) = \bar{y} + \frac{rf}{1-f} - \frac{(1-\tau)TC(fQ/T)}{(1-f)Q}. \quad (2)$$

The first term is the ordinary scale-neutral yield. The second term is the value of forfeited unvested rewards redistributed from exiters to the smaller pool of continuing holders. The last term is the adjustment-cost dilution borne by stayers. Thus, even when $y'(Q) = 0$, reward vesting can make the payoff from staying increase with aggregate exit.

The design objective is local: choose the least costly par-preserving mechanism that rules out run amplification at the stress state. The stability requirement is

$$\frac{\partial \Delta^{T,r}(f)}{\partial f} \geq 0. \quad (3)$$

Differentiating (2) shows that this constraint is equivalent to

$$r \geq r_{\min}(T) \equiv (1 - \tau) \frac{(1 - f)QC'(fQ/T) + TC(fQ/T)}{Q}. \quad (4)$$

The function $r_{\min}(T)$ is the minimum unvested reward needed to offset adjustment-cost dilution when redemptions are processed over T periods. A longer queue smooths a convex adjustment cost and lowers the required unvested reward, but it delays redemption for liquidity-shocked investors.

Let $\alpha > 0$ denote the liquidity cost of one additional period of exit delay. The protocol solves

$$\min_{T \geq 1, r \in [0, \bar{y}]} \left\{ TC\left(\frac{fQ}{T}\right) + fQ\alpha(T - 1) + fQr \right\} \quad \text{s.t.} \quad r \geq r_{\min}(T). \quad (5)$$

The three terms are, respectively, the real adjustment cost, the liquidity cost of delayed exit, and the welfare cost of making rewards unvested for liquidity-shocked investors.

Proposition 6 (Optimal reward vesting and exit queue). *At any interior optimum (T^*, r^*) of (5), the stability constraint binds:*

$$r^* = r_{\min}(T^*).$$

The optimal exit-processing time satisfies

$$fQ\alpha = \frac{fQ}{T^*} C'\left(\frac{fQ}{T^*}\right) - C\left(\frac{fQ}{T^*}\right) - fQr'_{\min}(T^*).$$

The proposition has a direct interpretation. The protocol makes just enough yield unvested to eliminate local run amplification. It does not maximize the unvested reward, because forfeiture is

costly for liquidity-shocked investors. It also does not maximize delay, because waiting is costly. The optimal queue equates the marginal liquidity cost of waiting with the combined benefit from smoother adjustment and the reduction in the required unvested reward.

Corollary 2 (Synthetic strategic substitutability). *If $r^* = r_{\min}(T^*) \leq \bar{y}$, then*

$$\frac{\partial \Delta^{T^*, r^*}(f)}{\partial f} = 0.$$

The mechanism exactly eliminates local run amplification. If the protocol chooses any feasible $r > r_{\min}(T^)$, then*

$$\frac{\partial \Delta^{T^*, r}(f)}{\partial f} > 0,$$

so staying becomes more attractive as aggregate exit rises.

This is synthetic strategic substitutability. Natural substitutability comes from the yield technology, $y'(Q) < 0$. Synthetic substitutability comes from the mechanism: exiters redeem at par but forfeit unvested rewards, and those rewards accrue to continuing holders. The mechanism therefore adapts the logic of redemption design to stablecoins without haircutting the base dollar claim. Related crypto designs already use locked or forfeitable rewards. For example, Chainlink Staking v0.2 distinguishes claimable rewards from locked rewards, and early unstaking can forfeit locked rewards that become available to other stakers ([Chainlink Labs, 2023](#)). Chainlink Rewards Season 1 similarly redistributes forfeited rewards to a loyalty pool for participants who do not early unlock ([Chainlink Labs, 2025](#)). The stablecoin-specific difference is the par constraint: vesting and forfeiture apply to the reward-bearing wrapper, not to the redemption value of the base stablecoin.

5.2 Adoption and the Medium-Scale Trap

The baseline model takes the scale of the protocol as given, thus studying redemption incentives conditional on an existing stock of claims. But the same force that stabilizes redemptions also has an ex ante implication for how large the protocol can become. This subsection develops that implication.

Natural scale relief comes from decreasing returns in the yield technology. When investors exit, the protocol becomes smaller, the underlying strategy becomes less crowded, and the continuation

yield for those who remain can rise. This stabilizes the redemption game. But adoption moves the protocol in the opposite direction. As more investors enter, the protocol becomes larger, the strategy becomes more crowded, and the yield that attracts investors can fall. The same decreasing-returns technology can therefore make the protocol more robust in distress and less scalable in normal times.

We capture this trade-off with a simple adoption stage. Before the redemption game, a mass M of potential investors decides whether to hold the yield-bearing stablecoin. Investor i has an idiosyncratic adoption cost χ_i , drawn from a continuous distribution F . This cost captures the investor's outside option, operational costs, regulatory constraints, or any other wedge that makes holding the token less attractive. Let $V(Q)$ denote the value of adopting when aggregate scale is Q , measured relative to the outside dollar asset. Investors adopt when

$$V(Q) \geq \chi_i.$$

Thus aggregate adoption must satisfy the fixed point

$$Q = MF(V(Q)). \quad (6)$$

This fixed-point structure is standard. The new force is in $V(Q)$: the value of adoption inherits the same scale-sensitive yield technology that governs the redemption game.

To keep the economics transparent, suppose potential adopters anticipate a constant exit fraction $\bar{f} \in [0, 1)$. The adoption value is

$$V(Q) = m(Q) + (1 - \bar{f})y((1 - \bar{f})Q) - \bar{f}k(\bar{f}Q) - (1 - \tau) \frac{C(\bar{f}Q)}{Q}. \quad (7)$$

The first term, $m(Q)$, is the payment or collateral convenience value of the token. This is the usual adoption benefit: a larger stablecoin may be more useful because it is more widely accepted, more liquid, or easier to use as collateral. The second term is the expected yield from the protocol's underlying strategy. Only investors who remain after the exit shock earn this yield, and the relevant scale of the strategy after exits is $(1 - \bar{f})Q$. The third term is the expected secondary-market discount borne by investors who need to exit. The last term is the expected incidence of balance-

sheet or trading adjustment costs. It is the same adjustment-cost object as in the redemption model, now evaluated from the perspective of an investor deciding whether to enter ex ante.

Differentiating (7) gives

$$V'(Q) = m'(Q) + (1 - \bar{f})^2 y'((1 - \bar{f})Q) - \bar{f}^2 k'(\bar{f}Q) - (1 - \tau) \frac{\bar{f}QC'(\bar{f}Q) - C(\bar{f}Q)}{Q^2}. \quad (8)$$

This derivative summarizes the adoption trade-off. The first term is positive when broader adoption raises payment or collateral convenience. The second term is negative in the crowded-trade regime $y' < 0$: larger scale compresses the yield generated by the strategy. The third term is negative when larger expected exit volume deepens the secondary-market discount. The last term is weakly negative under convex adjustment costs with $C(0) = 0$. Thus adoption can become self-limiting when yield compression and anticipated exit frictions dominate the marginal convenience benefit of scale.

The following result records the basic fixed-point property. It is not the main economic result of the subsection; it simply ensures that the adoption problem is well defined.

Proposition 7 (Adoption equilibrium). *If V is continuous on $[0, M]$, then an adoption equilibrium satisfying (6) exists. If*

$$M f_X(V(Q)) V'(Q) < 1 \quad \text{for all } Q \in [0, M],$$

then the equilibrium is unique.

The substantive implication concerns scale. Let Q^{pay} denote the scale required for broad payment use. This is not a technological boundary; it is a benchmark scale at which the stablecoin becomes a widely used payment or settlement instrument rather than a specialized yield-bearing product. A medium-scale trap arises when the protocol can sustain some adoption, but the adoption value falls before the protocol reaches Q^{pay} .

Proposition 8 (Medium-scale trap). *Let $Q^{\text{pay}} \in (0, M)$ be the scale required for broad payment use, and suppose F is strictly increasing on the relevant support. If $V'(Q) \leq 0$ for all $Q \geq Q^{\text{pay}}$ and*

$$V(Q^{\text{pay}}) < F^{-1}\left(\frac{Q^{\text{pay}}}{M}\right),$$

then no adoption equilibrium can support scale $Q \geq Q^{\text{pay}}$.

The condition has a direct graphical interpretation. The right-hand side of (6), $MF(V(Q))$, is the demand curve for the stablecoin at scale Q . If demand is already below the forty-five degree line at Q^{pay} , and if $V(Q)$ does not recover beyond that point, then demand remains below the level required to support broad payment scale. The protocol may still have an equilibrium at a smaller scale, but it cannot grow into a payment-scale instrument.

This mechanism differs from rail-congestion accounts of digital-money fragility. Here the negative scale effect does not come from blockchain settlement costs or gas fees. It comes from the protocol's own yield technology. Larger issuance requires a larger balance-sheet or derivatives strategy to generate the promised yield. If that strategy is capacity constrained, additional scale compresses the return available to adopters.

A linear example makes the trade-off transparent. Suppose

$$m(Q) = \bar{m} + \gamma Q, \quad y(Q) = \bar{y} - \phi Q,$$

with $\phi > 0$. The parameter γ captures the marginal convenience benefit of scale, while ϕ captures the strength of yield compression. If there is no anticipated exit, $\bar{f} = 0$, then

$$V(Q) = \bar{m} + \bar{y} + (\gamma - \phi)Q, \quad V'(Q) = \gamma - \phi.$$

Adoption value falls with scale if and only if $\phi > \gamma$. In words, the protocol becomes self-limiting when the marginal loss of yield from crowding the strategy exceeds the marginal payment benefit from broader adoption.

With anticipated exit, the same logic becomes stronger. From (8),

$$V'(Q) = \gamma - (1 - \bar{f})^2 \phi - \bar{f}^2 k'(\bar{f}Q) - (1 - \tau) \frac{\bar{f}Q C'(\bar{f}Q) - C(\bar{f}Q)}{Q^2}.$$

The secondary-market discount and adjustment-cost terms further reduce adoption value. Anticipated exit frictions therefore deepen the medium-scale trap: the protocol need not have very strong yield compression for adoption value to stop rising with scale.

The link to the exit model is immediate. In the redemption stage, under the same linear yield

technology, an exit fraction f raises the continuation yield by

$$y(Q(1 - f)) - y(Q) = \phi Q f.$$

A larger ϕ strengthens natural scale relief and therefore makes exits less amplifying. But in the adoption stage the same parameter lowers the value of entering:

$$\frac{\partial V(Q)}{\partial \phi} = -(1 - \bar{f})^2 Q < 0.$$

Thus the parameter that stabilizes redemptions also limits adoption. Yield-bearing stablecoins may be robust as medium-scale structured products precisely because redemptions de-crowd the underlying carry trade. But the same force can prevent them from scaling like payment stablecoins whose returns are not constrained by a crowded financial strategy.

6 Conclusion

Yield-bearing stablecoins combine two functions that are usually studied separately. They are meant to serve as stable transactional claims, but they also promise returns generated by an underlying balance-sheet or trading strategy. This paper shows that the interaction between these two functions is central for understanding fragility. In our model, investor exits affect not only redemption pressure but also the continuation value of remaining holders through the scale dependence of the yield technology. When returns decline with scale, redemptions can relieve crowding and create strategic substitutability rather than the standard run complementarity. When returns rise with scale, the same feedback amplifies exit incentives. The source of yield therefore changes the strategic structure of redemptions.

Our empirical analysis supports the natural scale-relief channel in the Ethena setting. Using daily data on USDe and sUSDe from August 2024 to January 2026, we find evidence of decreasing returns to scale, stronger redemptions when yield deteriorates, little evidence that current exits depress the future value of remaining claims, and a two-stage market response around delayed redemptions. These results suggest that the fragility of yield-bearing stablecoins cannot be understood solely through reserve quality, redemption access, or secondary-market liquidity. What matters equally is how the underlying yield is produced, how its profitability changes with scale, and how quickly the protocol can resize the positions that support it.

The extensions show how the same logic informs design and scalability. First, the stabilizing force need not come only from a naturally decreasing-return yield technology. Even when the underlying yield is closer to scale-neutral, a protocol can create synthetic strategic substitutability through reward vesting and exit queues. In the optimal par-preserving mechanism, exiting holders redeem at par but forfeit unvested rewards, which accrue to continuing holders. The protocol chooses the least costly combination of queue length and per-unit unvested reward needed to eliminate local run amplification. This mechanism adapts the economic logic of redemption design to stablecoins without directly haircutting the par redemption value.

Second, natural scale relief has a scalability cost. A yield technology with decreasing returns can soften run incentives because exits de-crowd the underlying strategy, but it also means that adoption compresses the yield that attracts investors in the first place. The extension formalizes

this trade-off as a medium-scale trap: yield-bearing stablecoins may be robust as medium-scale structured products precisely because the underlying strategy is capacity constrained, but the same force can prevent them from scaling like broad payment instruments. More scalable yield-bearing stablecoins may therefore need to rely less on natural scale relief and more on explicit reward and redemption design.

Yield-bearing stablecoins should be evaluated not only by whether they pay yield, but by the balance sheet or trading technology that generates that yield and by the mechanism that governs exit. Tokenized deposits inherit fragility from balance-sheet maturity transformation; yield-bearing stablecoins inherit fragility from strategy capacity, margin, market liquidity, and reward design. The design of on-chain dollar instruments will therefore shape not only their private attractiveness to investors, but also the resilience of the markets in which they operate.

References

- Adams, A. and Ibert, M. (2022). Runs on algorithmic stablecoins: Evidence from iron, titan, and steel. *Working Paper*.
- Aldasoro, I., Ahmed, R., and Duley, C. (2023). Par for the course: Public information and stablecoin runs. *SSRN Working Paper*.
- Allen, F. and Gale, D. (1998). Optimal financial crises. *Journal of Finance*, 53:1245–1284.
- Berk, J. B. and Green, R. C. (2004). Mutual fund flows and performance in rational markets. *Journal of Political Economy*, 112(6):1269–1295.
- Bernardo, A. E. and Welch, I. (2004). Liquidity and financial market runs. *Quarterly Journal of Economics*, 119:135–158.
- Bertsch, C. (2023). Stablecoins: Adoption and fragility. *Working Paper*.
- Capponi, A., Glasserman, P., and Weber, M. (2020). Swing pricing for mutual funds: Breaking the feedback loop between fire sales and fund redemptions. *Management Science*, 66(8):3581–3602.
- Carhart, M. M., Kaniel, R., Musto, D. K., and Reed, A. V. (2002). Leaning for the tape: Evidence of gaming behavior in equity mutual funds. *The Journal of Finance*, 57(2):661–693.
- Chainlink Labs (2023). Chainlink staking v0.2 overview. Chainlink Blog. Accessed 2026.
- Chainlink Labs (2025). Introducing chainlink rewards season 1. Chainlink Blog. Accessed 2026.
- Chan, L. K. C. and Lakonishok, J. (1995). The behavior of stock prices around institutional trades. *The Journal of Finance*, 50(4):1147–1174.
- Chen, J., Hong, H., Huang, M., and Kubik, J. D. (2004). Does fund size erode mutual fund performance? the role of liquidity and organization. *American Economic Review*, 94(5):1276–1302.
- Christin, N., Routledge, B., Soska, K., and Zetlin-Jones, A. (2023). The crypto carry trade. Working paper.

- d'Avernas, A., Maurin, V., and Vandeweyer, Q. (2022). Can stablecoins be stable? Working Paper 2022-131, Becker Friedman Institute for Economics.
- Diamond, D. W. and Dybvig, P. H. (1983). Bank runs, deposit insurance, and liquidity. *Journal of Political Economy*, 91:401–419.
- Duffie, D. (2010). Presidential address: Asset price dynamics with slow moving capital. *The Journal of Finance*, 65(4):1237–1267.
- European Parliament and Council of the European Union (2023). Regulation (eu) 2023/1114 of the european parliament and of the council of 31 may 2023 on markets in crypto-assets. Official Journal of the European Union. Markets in Crypto-Assets Regulation (MiCA).
- Forsberg, D., Gallagher, D. R., and Warren, G. J. (2022). Capacity constraints in hedge funds: The relation between fund performance and cohort size. *Financial Analysts Journal*, 78(2):57–77.
- Gabaix, X. and Koijen, R. S. J. (2021). In search of the origins of financial fluctuations: The inelastic markets hypothesis. NBER Working Paper No. 28967.
- Gârleanu, N. and Pedersen, L. H. (2011). Margin-based asset pricing and deviations from the law of one price. *Review of Financial Studies*, 24(6):1980–2022.
- Goldstein, I. and Pauzner, A. (2005). Demand deposit contracts and the probability of bank runs. *Journal of Finance*, 60:1293–1328.
- Gorton, G. B., Klee, E. C., Ross, C. P., Ross, S. Y., and Vardoulakis, A. P. (2025). Leverage and stablecoin pegs. Available at SSRN. SSRN working paper, last revised April 3, 2025.
- Gorton, G. B., Ross, C. P., and Ross, S. Y. (2022). Making money. Working Paper 29710, National Bureau of Economic Research.
- Gromb, D. and Vayanos, D. (2002). Equilibrium and welfare in markets with financially constrained arbitrageurs. *Journal of Financial Economics*, 66(2-3):361–407.
- Harris, L. and Gurel, E. (1986). Price and volume effects associated with changes in the s&p 500 list: New evidence for the existence of price pressures. *Journal of Finance*, 41(4):815–829.

- He, Z., Manela, A., Ross, P., and Von Wachter, J. (2024). Fundamentals of perpetual futures. Working paper, arXiv:2212.06888.
- Jin, D., Kacperczyk, M., Kahraman, B., and Suntheim, F. (2022). Swing pricing and fragility in open-end mutual funds. *Review of Financial Studies*, 35(1):1–50.
- Kacperczyk, M. and Schnabl, P. (2013). How safe are money market funds? *Quarterly Journal of Economics*, 128:1073–1122.
- Keim, D. B. and Madhavan, A. (1997). Transactions costs and investment style: An inter-exchange analysis of institutional equity trades. *Journal of Financial Economics*, 46(3):265–292.
- Lewrick, U. and Schanz, J. (2023). Towards a macroprudential framework for investment funds: Swing pricing and investor redemptions. *International Journal of Central Banking*, 19(3):229–267.
- Li, L., Li, Y., Macchiavelli, M., and Zhou, X. (2021). Liquidity restrictions, runs, and central bank interventions: Evidence from money market funds. *The Review of Financial Studies*, 34(11):5402–5437.
- Li, Y. and Mayer, S. (2021). Money creation in decentralized finance: A dynamic model of stablecoin and crypto shadow banking. Working Paper 9260, CESifo.
- Liu, J., Makarov, I., and Schoar, A. (2023). Anatomy of a run: The terra luna crash. *Working Paper*.
- Lou, D. (2012). A flow-based explanation for return predictability. *Review of Financial Studies*, 25(12):3457–3489.
- Lyons, R. and Viswanath-Natraj, G. (2021). What keeps stablecoins stable? *Working Paper*.
- Lyons, R. K. and Viswanath-Natraj, G. (2023). What keeps stablecoins stable? *Journal of International Money and Finance*, 131:102777.
- Ma, Y., Zeng, Y., and Zhang, A. L. (2025). Stablecoin runs and the centralization of arbitrage. Technical report, National Bureau of Economic Research.

- Matta, R., Oostdam, R., and Perotti, E. (2026). Sand in the wheels: Addressing runs through contingent fees. Working paper.
- McLemore, P. (2019). Do mutual funds have decreasing returns to scale? evidence from fund mergers. *Journal of Financial and Quantitative Analysis*, 54(4):1683–1711.
- Mitchell, M., Pedersen, L. H., and Pulvino, T. (2007). Slow moving capital. *American Economic Review*, 97(2):215–220.
- Morris, S. and Shin, H. S. (1998). Unique equilibrium in a model of self-fulfilling currency attacks. *American Economic Review*, 88(3):587–597.
- Naik, N. Y., Ramadorai, T., and Strömquist, M. (2007). Capacity constraints and hedge fund strategy returns. *European Financial Management*, 13(2):239–256.
- Neuder, M., Pai, M. M., and Resnick, M. (2024). Optimizing exit queues for proof-of-stake blockchains: A mechanism design approach. In *6th Conference on Advances in Financial Technologies (AFT 2024)*, volume 316 of *Leibniz International Proceedings in Informatics (LIPIcs)*, pages 20:1–20:22. Schloss Dagstuhl – Leibniz-Zentrum für Informatik.
- Parlatore, C. (2016). Fragility in money market funds: Sponsor support and regulation. *Journal of Financial Economics*, 121:595–623.
- Pástor, L., Stambaugh, R. F., and Taylor, L. A. (2015). Scale and skill in active management. *Journal of Financial Economics*, 116(1):23–45.
- Ramadorai, T. (2013). Capacity constraints, investor information, and hedge fund returns. *Journal of Financial Economics*, 107(2):401–416.
- Routledge, B. and Zetlin-Jones, A. (2022). Currency stability using blockchain technology. *Journal of Economic Dynamics and Control*, 142:104155.
- Schmidt, L., Timmermann, A., and Wermers, R. (2016). Runs on money market mutual funds. *American Economic Review*, 106:2625–2657.
- Shleifer, A. (1986). Do demand curves for stocks slope down? *Journal of Finance*, 41(3):579–590.

Shleifer, A. and Vishny, R. W. (1997). The limits of arbitrage. *Journal of Finance*, 52(1):35–55.

Skeie, D. (2021). Digital currency runs. Working paper, Warwick Business School.

Stein, J. C. (2009). Presidential address: Sophisticated investors and market efficiency. *The Journal of Finance*, 64(4):1517–1548.

Sunderam, A. (2015). Money creation and the shadow banking system. *Review of Financial Studies*, 28(4):939–977.

Uhlig, H. (2022). A luna-tic stablecoin crash. Working Paper 30256, National Bureau of Economic Research.

United States Congress (2025). Guiding and establishing national innovation for u.s. stablecoins act of 2025. GENIUS Act.

Wurgler, J. and Zhuravskaya, E. (2002). Does arbitrage flatten demand curves for stocks? *Journal of Business*, 75(4):583–608.

A Appendix: Proofs

A.1 Proof of Proposition 1

The payoff from selling in the secondary market is $W_L(f) = 1 - k(fQ)$ and the payoff from staying is

$$W_S(f) = 1 + y(Q(1 - f)) - \frac{(1 - \tau)C(fQ)}{(1 - f)Q}.$$

The incentive to stay is therefore

$$\Delta(f) = W_S(f) - W_L(f) = k(fQ) + y(Q(1 - f)) - \frac{(1 - \tau)C(fQ)}{(1 - f)Q}.$$

We differentiate this expression with respect to f .

The derivative of the secondary-market discount term follows from the chain rule, since the argument fQ is linear in f with slope Q :

$$\frac{d}{df} k(fQ) = Q k'(fQ).$$

The derivative of the yield term likewise follows from the chain rule, since $Q(1 - f)$ has slope $-Q$ with respect to f :

$$\frac{d}{df} y(Q(1 - f)) = y'(Q(1 - f)) \cdot (-Q) = -Q y'(Q(1 - f)).$$

The derivative of the pool-dilution term follows from the quotient rule, treating numerator $(1 - \tau)C(fQ)$ and denominator $(1 - f)Q$ each as functions of f :

$$\begin{aligned} \frac{d}{df} \left(\frac{(1 - \tau)C(fQ)}{(1 - f)Q} \right) &= \frac{(1 - \tau)}{Q} \cdot \frac{d}{df} \left(\frac{C(fQ)}{1 - f} \right) \\ &= \frac{(1 - \tau)}{Q} \cdot \frac{Q C'(fQ) (1 - f) + C(fQ)}{(1 - f)^2} \\ &= (1 - \tau) \frac{(1 - f)Q C'(fQ) + C(fQ)}{(1 - f)^2 Q}. \end{aligned}$$

Collecting all three terms yields

$$\Delta'(f) = Q k'(fQ) - Q y'(Q(1-f)) - (1-\tau) \frac{(1-f)Q C'(fQ) + C(fQ)}{(1-f)^2 Q}.$$

When $\Delta'(f) < 0$, a higher aggregate exit rate reduces the relative payoff to staying, generating strategic complementarity among sellers. When $\Delta'(f) > 0$, a higher exit rate makes staying more attractive, generating strategic substitutability. The first term $Q k'(fQ) > 0$ is always positive and stabilizing, with magnitude increasing in protocol scale Q . The second term is positive when $y'(Q) < 0$ and negative when $y'(Q) > 0$. The third term is always negative and destabilizing, and likewise increasing in magnitude with Q because a given exit fraction f corresponds to a larger redeemed notional the larger the outstanding liability. $\square$

A.2 Proof of Proposition 2

The global-games structure follows [Morris and Shin \(1998\)](#), applied to the retail investor's decision between selling in the secondary market and remaining invested. Each investor i observes a private signal $s_i = \theta + \varepsilon_i$, where ε_i is independently distributed with full support.

With the payoffs established above, the incentive difference is

$$\Delta(f, \theta) = k(fQ) + \bar{y}(Q(1-f)) + \theta - \frac{(1-\tau)C(fQ)}{(1-f)Q},$$

which is strictly increasing in θ for any fixed f , since θ enters additively with a coefficient of one. Accordingly, the payoff from selling relative to staying is strictly decreasing in θ : higher fundamentals raise the continuation yield and make staying strictly more attractive.

This monotonicity ensures that the dominance regions are non-empty. For θ sufficiently low, the continuation yield is so depressed that selling dominates staying for all investors regardless of their beliefs about others. For θ sufficiently high, staying dominates selling unconditionally.

Iterated elimination of strictly dominated strategies under small and diffuse signal noise yields a unique rationalizable outcome that converges to a threshold strategy: investor i sells if and only if $s_i < s^*$, where s^* is the unique cutoff satisfying the indifference condition $\Delta(f^*(s^*), s^*) = 0$ and $f^*(s^*)$ is the exit fraction generated by threshold s^* under the law of large numbers. $\square$

A.3 Proof of Proposition 3

The equilibrium threshold θ^* satisfies $\Delta(f^*(\theta^*), \theta^*) = 0$, where

$$\Delta(f, \theta) = k(fQ) + \bar{y}(Q(1 - f)) + \theta - \frac{(1 - \tau)C(fQ)}{(1 - f)Q}.$$

Let $\alpha \equiv -y'(Q) > 0$ parameterize the magnitude of the scale sensitivity. A higher α means that each unit reduction in scale generates a larger increase in yield, so $\Delta(f, \theta)$ rises for all (f, θ) as α increases. At the prevailing equilibrium threshold θ^* , the incentive to stay is now positive, so θ^* must fall to restore indifference.

Applying the implicit function theorem to $\Delta(f^*(\theta^*), \theta^*) = 0$:

$$\frac{\partial \theta^*}{\partial \alpha} = -\frac{\partial \Delta / \partial \alpha}{\partial \Delta / \partial \theta^*}.$$

The denominator equals one, since θ enters Δ only additively. The numerator $\partial \Delta / \partial \alpha$ is negative, because a higher α raises Δ and reduces the threshold level of θ required for indifference. Therefore $\partial \theta^* / \partial \alpha < 0$: stronger yield relief from scale reductions lowers the run threshold. $\square$

A.4 Proof of Corollary 1

Under the linear specification $y(Q) = \bar{y} - \phi Q$ with $\phi > 0$, the yield available to remaining investors after exit fraction f is

$$y(Q(1 - f)) = \bar{y} - \phi Q(1 - f).$$

This expression is strictly decreasing in Q for any fixed f , since $\phi > 0$. An increase in outstanding supply Q through minting therefore directly reduces the yield available to all investors through the crowding of the underlying derivatives trade. This result concerns the yield technology alone and is independent of the secondary-market discount $k(fQ)$ and the adjustment cost $C(fQ)$. Conversely, a reduction in Q through arbitrageur redemption relieves crowding and restores yield. $\square$

A.5 Proof of Proposition 4

The result follows from Jensen's inequality applied to the strictly convex cost function C . Since C is strictly convex, for any $x \neq z$ and $\lambda \in (0, 1)$,

$$C(\lambda x + (1 - \lambda)z) < \lambda C(x) + (1 - \lambda)C(z).$$

Consider total redeemed notional fQ processed by arbitrageurs uniformly over T cooldown subperiods, with each subperiod handling a tranche of notional size fQ/T . The total protocol-side adjustment cost of the staged implementation is $T \cdot C(fQ/T)$. By the strict convexity of C ,

$$C(fQ) = C\left(\sum_{t=1}^T \frac{fQ}{T}\right) > T C\left(\frac{fQ}{T}\right),$$

where the inequality holds because the average cost of processing T tranches of notional fQ/T is strictly less than the cost of processing the full redeemed notional fQ in a single redemption. Therefore $T C(fQ/T) < C(fQ)$: spreading the redeemed notional over the cooldown subperiods reduces total protocol-side adjustment costs. Because the fraction $1 - \tau$ of these costs is borne by remaining investors, the cooldown also reduces the dilution of the continuing pool. $\square$

A.6 Proof of Proposition 5

Under the cooldown mechanism, retail investors submit unstake requests at announcement date $t = 0$. Arbitrageurs then process the associated redeemed notional over T cooldown subperiods beginning at $t = 1$, implementing a tranche of notional size fQ/T in each subperiod.

At $t = 0$, the aggregate volume of unstake requests becomes observable to all market participants. This announcement is informative about the forthcoming redeemed notional, the expected path of protocol scale, and the anticipated trajectory of the funding rate basis. Under rational expectations, secondary-market prices and the associated discount $k(fQ)$ may adjust at $t = 0$ even though no arbitrageur redemptions have yet been executed and no underlying positions have been unwound.

A second and distinct adjustment occurs at dates $t = 1, \dots, T$ as arbitrageurs redeem each

tranche of notional fQ/T and the protocol closes the corresponding underlying positions. These execution-stage adjustments reflect the realized impact of position unwinding on derivatives market prices and the funding rate basis. They need not be fully anticipated at $t = 0$ because the mapping from the announced unstake volume to the realized redemption schedule may be imperfect.

Because the announcement stage and the execution stage are separated in time by the cooldown, market quantities related to exit activity exhibit a predictable two-stage pattern: an initial response at the time of observable unstake requests, followed by a second response at the time of actual redemption execution in tranches of notional fQ/T . □

B Appendix: Tables

Table A1: DESCRIPTIVE STATISTICS

Daily observations, August 1, 2024 – January 31, 2026 ($N = 549$).

	Mean	Std. Dev.	p25	Median	p75	N
<i>Panel A: Yield and NAV</i>						
NAV log return (daily)	0.0002	0.0018	-0.0001	0.0004	0.0005	549
NAV level (USDe/sUSDe)	1.090	0.036	1.064	1.085	1.114	549
Daily rewards (000s USDe)	680	478	340	563	902	549
<i>Panel B: Protocol flows ($N = 443$, supply-valid subsample)</i>						
USDe supply (\$B)	3.98	2.88	2.17	2.77	4.63	443
Minted (M USDe/day)	29.6	85.3	0.0	4.3	27.6	443
Burned (M USDe/day)	24.6	90.2	0.0	3.8	19.4	443
Net issuance (M USDe/day)	6.1	115.4	-9.4	0.4	8.2	443
Unstake rate (fraction of AUM)	0.138	2.699	0.000	0.001	0.007	443
Cooldown start rate (fraction)	0.045	0.805	0.000	0.001	0.006	443
Cooldown starts per day	36.2	34.0	17.0	25.0	42.0	549
<i>Panel C: Market conditions</i>						
ETH spot price (\$)	2,847	442	2,115	2,763	4,109	549
Basis, mark vs. spot (bps)	-3.49	2.96	-4.78	-3.65	-3.32	549
8-hr funding rate (bps)	0.64	2.87	0.00	0.50	1.00	548
Open interest (M USDT)	836	229	402	731	1,021	549
USDe peg deviation (bps)	-2.3	22.8	-14.6	-0.8	8.2	538
Abs. peg deviation (bps)	17.5	14.8	5.2	12.8	24.6	538

Notes. Panel B covers the supply-valid subsample available from November 17, 2024 onward ($N = 443$); all other variables span the full 549-day sample. Flow rates in Panel B are expressed as a fraction of contemporaneous AUM. The basis is $(F_t - S_t)/S_t \times 10,000$, where F_t is the Binance ETHUSDT perpetual mark price and S_t is ETH spot. NAV is reconstructed from `RewardsReceived` on-chain events on the `StakedUSDeV2` contract; the daily return is the log change in this reconstructed series. Peg deviation is $(P_t - 1.00) \times 10,000$ in basis points, where P_t is the volume-weighted Curve Finance USDe-USDC price.

Table A2: SCALE AND YIELD

Panel A reports OLS regressions of annualized sUSDe yield on log lagged supply. Panel B sorts supply-valid days into quintiles of lagged supply and reports mean yield per quintile.

<i>Dependent variable: annualized yield (%)</i>				
	(1) Daily	(2) 7-day	(3) 30-day	(4) Log NAV
$\log Q_{t-1}$	11.50** (2.09)	4.79 (0.83)	-18.57*** (-2.70)	-0.0297*** (-2.97)
<i>Controls</i>				
Phase dummies	Yes	Yes	Yes	Yes
Lag basis	Yes	Yes	Yes	Yes
N	441	440	429	441
R^2	0.026	0.043	0.115	0.491
<i>Panel B: Non-parametric evidence — supply quintiles</i>				
Quintile	Mean supply (\$B)	30-day yield (%)	NAV level	N
Q1 (lowest)	1.57	15.68	1.119	88
Q2	2.33	7.91	1.084	88
Q3	2.80	-0.42	1.083	88
Q4	4.07	13.65	1.124	88
Q5 (highest)	9.24	-3.51	1.077	88
Spearman $\rho = -0.111$ ($p = 0.021$)				

Notes. Panel A regresses annualized sUSDe yield on log outstanding USDe supply lagged one day ($\log Q_{t-1}$), with the dependent variable smoothed at the daily, 7-day rolling, 30-day rolling, and log NAV level horizons in columns (1)–(4) respectively. The positive coefficient in column (1) reflects a day-boundary artifact from the 8-hour reward-vesting schedule, in which the day of reward attribution varies mechanically across the UTC midnight boundary; this noise is absorbed at the 30-day horizon, where the true negative relationship is recovered. All specifications include Phase 2 and Phase 3 dummies and the lagged mark-minus-spot basis. Standard errors are Newey–West with ten lags; t -statistics in parentheses. Panel B sorts the 443 supply-valid days into quintiles of $\log Q_{t-1}$ and reports mean 30-day annualized yield and mean NAV level per quintile. The non-monotonicity between Q3 and Q4 reflects the initial portion of the GENIUS Act surge (Table A3), before yield compression fully materialized. * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

Table A3: THE GENIUS ACT AS A NATURAL EXPERIMENT

Two-stage least squares using the GENIUS Act signing as an instrument for USDe supply. Dependent variable in the second stage is the 30-day rolling annualized sUSDe yield. Sample: January 1, 2025 – January 31, 2026 ($N = 396$).

	<i>Panel A: First stage</i> $\log Q_t$	<i>Panel B: Second stage</i> 30-day yield (%)
Post-GENIUS (Z_t)	0.954*** (8.36)	
$\widehat{\log Q_t}$ (IV)		−11.957* (−1.80)
Phase 3	−0.024 (−0.42)	5.204 (0.61)
Lag basis	−0.012 (−0.84)	−1.569 (−1.49)
Constant	0.736*** (8.95)	14.320 (1.44)
<i>Diagnostics</i>		
N	396	396
R^2	0.618	0.017
First-stage F	69.9	
<i>OLS benchmark (same sample)</i>		
$\log Q_{t-1}$		−18.99*** (−2.71)

Notes. The instrument $Z_t = \mathbf{1}[t \geq \text{July 18, 2025}]$ indicates post-signing observations. Panel A confirms the supply surge is strongly predicted by the signing date: post-signing supply averages $e^{0.95} \approx 2.6$ times its pre-signing level, with a first-stage F -statistic of 69.9 that comfortably exceeds the weak-instrument threshold. Panel B regresses 30-day rolling annualized yield on instrumented log supply; the coefficient implies a doubling of AUM reduces annualized yield by approximately 12 percentage points. The OLS benchmark (−19.0***) exceeds the IV estimate in absolute value, consistent with mild upward OLS bias from yield-pull: in the full sample, supply growth tends to coincide with favorable carry environments as investors are attracted by returns, biasing the OLS coefficient away from zero. Exclusion restriction checks are reported in the text. All specifications control for Phase 3 dummy and lagged mark-minus-spot basis. Standard errors are Newey–West with ten lags; t -statistics in parentheses. * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

Table A4: EXIT FLOWS AND NAV DYNAMICS

Panel A reports the first stage of the cooldown IV. Panels B and C report OLS and IV estimates from regressing log NAV return at horizon k on exit flows. First-stage F -statistic on the excluded instrument is 9,461.

Panel A: First stage — cooldown initiations predicting unstakes

	<i>Dep. var.: unstake rate_t</i>		
	(1)	(2)	(3)
Cooldown rate _{$t-7$}	0.8693*** (20.82)	0.8658*** (20.80)	0.8653*** (20.67)
Phase dummies	No	Yes	Yes
Lag NAV, lag basis	No	No	Yes
N	434	434	434
R^2	0.930	0.931	0.931
First-stage F	9,461		

Panel B: OLS — exit flows predicting future log NAV return

Horizon k	1 day	3 days	5 days	7 days	9 days
Burns: $\hat{\beta}$	0.00000	0.00000	0.00000	-0.00000	0.00002***
t	(0.40)	(1.25)	(1.53)	(-0.12)	(13.24)
Unstakes: $\hat{\beta}$	0.00000	0.00001*	0.00000	-0.00000	0.00003***
t	(0.38)	(1.66)	(1.62)	(-0.15)	(12.84)
N	442	440	438	436	434

Panel C: IV — instrumented unstakes predicting future log NAV return

Horizon k	1 day	3 days	5 days	7 days	9 days
$\hat{\beta}_{IV}$	0.00285	0.00554*	0.00681***	0.00168	0.00486
t	(1.16)	(1.70)	(2.63)	(0.68)	(1.33)
N	433	431	429	427	425

Notes. Panel A reports the first stage of the cooldown instrumental variables strategy; the dependent variable is the unstake rate (unstaked USDe divided by outstanding AUM) on day t and the instrument is the cooldown initiation rate on day $t - 7$. The seven-day gap reflects the mandatory cooldown period fixed by the protocol's smart contract on November 20, 2023. The same-day correlation between the cooldown rate and the unstake rate is only 0.069, confirming the instrument operates exclusively through the predetermined lag and not through contemporaneous common shocks. The first-stage F -statistic of 9,461 rules out weak-instrument concerns. Panels B and C report horizon-by-horizon estimates from equation (1), where the dependent variable is the log NAV return at horizon k days ahead. The fire-sale mechanism requires $\hat{\beta} < 0$; no flow measure and no horizon produces this. The positive coefficients at $k = 9$ reflect the 8-hour reward-vesting schedule, which creates mild positive autocorrelation at that interval and is unrelated to exit flows. Panel C instruments the unstake rate with the cooldown initiation rate seven days prior; IV coefficients are positive at all horizons, opposite to the fire-sale prediction and ruling out attenuation bias. All specifications in Panels B and C include Phase 2 and Phase 3 dummies, lagged log NAV return, and lagged mark-minus-spot basis. Standard errors are Newey–West with five lags; t -statistics in parentheses. * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

Table A5: PHASE HETEROGENEITY AND REVERSE CAUSALITY

Panels A and B split the fire-sale test by collateral regime. Panel C reverses the regression to test whether investors respond to yield signals.

Panel A: Burns predicting future log NAV return, by phase

Horizon k	Phase 1 (ETH/BTC)		Phase 2 (Mixed)		Phase 3 (Stable)	
	$\hat{\beta}$	t	$\hat{\beta}$	t	$\hat{\beta}$	t
1	0.00000	(1.10)	0.00726	(1.08)	0.01317	(1.25)
3	0.00000	(0.70)	0.00358	(0.74)	0.01158*	(1.94)
5	0.00000***	(4.92)	-0.00136	(-0.24)	0.01546**	(2.05)
7	0.00000**	(2.14)	-0.00161	(-0.28)	0.00503	(0.49)

Panel B: Unstakes predicting future log NAV return, by phase

Horizon k	Phase 1 (ETH/BTC)		Phase 2 (Mixed)		Phase 3 (Stable)	
	$\hat{\beta}$	t	$\hat{\beta}$	t	$\hat{\beta}$	t
1	0.00000	(1.03)	0.00267	(1.19)	0.01293	(0.82)
3	0.00000	(1.21)	0.00558*	(1.84)	0.01539	(1.35)
5	0.00000***	(5.03)	0.00626**	(2.46)	0.02507*	(1.71)
7	0.00000**	(2.03)	0.00099	(0.33)	0.01143	(1.35)

Panel C: Basis predicting future cooldown initiations (reverse causality)

Dep. var.: cooldown initiation rate at horizon k

Horizon k	$\hat{\beta}$	t	N
1	-0.03998	(-0.94)	442
3	-0.00030*	(-1.66)	441
5	-0.00043**	(-2.21)	441
7	-0.00045***	(-3.05)	441
9	-0.00044**	(-2.28)	441
14	-0.00002	(-0.07)	441

Notes. Panels A and B report horizon-by-horizon estimates from equation (1) estimated separately within each collateral phase. Phase 1 covers August 1 – December 31, 2024 ($N \leq 153$), the ETH- and BTC-dominant period in which the protocol's perp positions are in the most liquid markets and adjustment costs are lowest; the fire-sale mechanism, if operative, should be strongest in this phase. Phase 2 covers January 1 – June 30, 2025 ($N \leq 181$). Phase 3 covers July 1, 2025 – January 31, 2026 ($N \leq 215$). No phase produces a negative coefficient at the horizons where the fire-sale mechanism should operate. Positive coefficients in Phase 3 reflect yield-environment correlation: large redemption events in the stablecoin-dominant phase coincide with periods when the protocol is actively growing. Controls include lagged log NAV return and lagged basis. Standard errors are Newey–West with three lags within phase. Panel C regresses the cooldown initiation rate k days ahead on the current mark-minus-spot basis in basis points. A negative coefficient means a lower basis today predicts more exit initiations in the following week. The effect peaks at $k = 7$ ($\hat{\beta} = -0.00045$, $t = -3.05$), consistent with investors responding to yield signals over a one-week horizon rather than running strategically to beat one another. Controls include phase dummies and lagged log NAV return. Standard errors are Newey–West with five lags. * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

Table A6: EXIT BEHAVIOR AND REGIME DEPENDENCE

Dependent variables are daily burns (columns 1–2) and net issuance (column 3), all in millions of USDe. The sample is 549 daily observations, August 1, 2024 – January 31, 2026.

	(1) Burns	(2) Burns	(3) Net issuance
Basis	−0.90* (−1.50)	−0.90 (−1.50)	
Basis × 1 [Basis < 0]		−9.42** (−2.49)	
1 [Basis < 0]		−36.24** (−2.23)	
Funding rate			48.13*** (4.89)
Funding rate × 1 [Funding < 0]			−6.92 (−0.54)
1 [Funding < 0]			23.69*** (2.99)
Volatility	2.76 (0.95)	2.76 (0.95)	−0.10 (−0.06)
USDe supply	5.78** (2.36)	5.78** (2.36)	
Constant	−9.37 (−0.70)	−9.37 (−0.70)	−26.52*** (−2.71)
<i>N</i>	549	549	549
<i>R</i> ²	0.032	0.194	0.113

Notes. All flow variables are in millions of USDe. The basis is the mark-minus-spot perpetual spread in basis points; the funding rate is the daily average 8-hour rate in basis points; volatility is the seven-day rolling standard deviation of daily ETH spot returns; USDe supply is in billions of dollars. Column (1) documents a baseline negative relationship between the basis and daily burns: a one basis point decline in the basis is associated with approximately 0.9 million USDe of incremental daily redemptions. Column (2) introduces the indicator **1**[Basis < 0] and its interaction with the basis. The interaction coefficient of −9.42 ($t = -2.49$) implies that exit sensitivity is approximately 11.4 times larger when the basis is below zero, consistent with the model’s prediction that marginal investors become more responsive to further yield compression when carry is already near zero. Despite this amplification, the absence of fire-sale dynamics in Table A4 holds throughout, indicating that the amplification operates through individual optimization rather than through the strategic complementarity channel. Column (3) estimates the entry-side counterpart; the asymmetry for below-zero funding rates is statistically indistinguishable from zero, suggesting regime dependence is sharper on the exit side. Standard errors are Newey–West with five lags; t -statistics in parentheses. * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

C Appendix: Extensions

C.1 Reward Vesting and Exit Queues

Proof of Proposition 6. Under a design (T, r) , the incentive to stay is

$$\Delta^{T,r}(f) = \bar{y} + \frac{rf}{1-f} - \frac{(1-\tau)TC(fQ/T)}{(1-f)Q}.$$

Holding T fixed and differentiating with respect to f gives

$$\frac{\partial \Delta^{T,r}(f)}{\partial f} = \frac{r}{(1-f)^2} - (1-\tau) \frac{(1-f)QC'(fQ/T) + TC(fQ/T)}{(1-f)^2Q}.$$

Therefore the no-run-amplification condition $\partial \Delta^{T,r}(f)/\partial f \geq 0$ is equivalent to

$$r \geq r_{\min}(T), \quad r_{\min}(T) = (1-\tau) \frac{(1-f)QC'(fQ/T) + TC(fQ/T)}{Q}.$$

The protocol minimizes

$$TC\left(\frac{fQ}{T}\right) + fQ\alpha(T-1) + fQr$$

subject to $r \geq r_{\min}(T)$. The objective is strictly increasing in r . Thus, at any interior optimum, the constraint must bind; otherwise the protocol could reduce r , remain feasible, and lower the objective. Hence

$$r^* = r_{\min}(T^*).$$

Substituting the binding constraint gives the one-dimensional objective

$$G(T) = TC\left(\frac{fQ}{T}\right) + fQ\alpha(T-1) + fQr_{\min}(T).$$

Let $x(T) = fQ/T$. Since $x'(T) = -x(T)/T$,

$$\frac{d}{dT} [TC(x(T))] = C(x(T)) - x(T)C'(x(T)).$$

Thus

$$G'(T) = C(x(T)) - x(T)C'(x(T)) + fQ\alpha + fQr'_{\min}(T).$$

At an interior optimum $T^* > 1$, $G'(T^*) = 0$. Substituting $x(T^*) = fQ/T^*$ and rearranging yields

$$fQ\alpha = \frac{fQ}{T^*}C'\left(\frac{fQ}{T^*}\right) - C\left(\frac{fQ}{T^*}\right) - fQr'_{\min}(T^*),$$

as claimed. □

Proof of Corollary 2. The proof above implies

$$\frac{\partial \Delta^{T,r}(f)}{\partial f} = \frac{1}{(1-f)^2} [r - r_{\min}(T)].$$

At the optimal mechanism, $r^* = r_{\min}(T^*)$, provided this value is feasible. Therefore

$$\frac{\partial \Delta^{T^*,r^*}(f)}{\partial f} = 0.$$

If instead $r > r_{\min}(T^*)$, then

$$\frac{\partial \Delta^{T^*,r}(f)}{\partial f} > 0.$$

The incentive to stay is then strictly increasing in aggregate exit, even though the underlying yield is scale-neutral. This is synthetic strategic substitutability. □

Interior queue length. The proposition characterizes any interior optimum. A simple sufficient condition for an interior optimum is $G'(1) < 0$ and $\lim_{T \rightarrow \infty} G'(T) > 0$. The first inequality says that, starting from immediate redemption, the marginal benefit of smoothing redemptions and lowering the required unvested reward exceeds the marginal liquidity cost of delay. The second says that very long queues are eventually unattractive. Under regularity conditions such as $C(0) = 0$, finite $C'(0)$, and $r'_{\min}(T) \rightarrow 0$, we have

$$\lim_{T \rightarrow \infty} G'(T) = fQ\alpha > 0.$$

If G is strictly convex on $T \geq 1$, the interior optimum is unique.

C.2 Adoption and the Medium-Scale Trap

Proof of Proposition 7. Define $\Phi(Q) = MF(V(Q))$. Since V and F are continuous, Φ is continuous. Because F takes values in $[0, 1]$, Φ maps $[0, M]$ into itself. Hence, by the intermediate value theorem applied to $\Phi(Q) - Q$, there exists $Q^* \in [0, M]$ such that $Q^* = \Phi(Q^*)$.

For uniqueness, note that

$$\Phi'(Q) = M f_\chi(V(Q)) V'(Q).$$

If $M f_\chi(V(Q)) V'(Q) < 1$ for all $Q \in [0, M]$, then $\Phi(Q) - Q$ is strictly decreasing. It can cross zero at most once. The adoption equilibrium is therefore unique. $\square$

Proof of Proposition 8. Since $V'(Q) \leq 0$ for all $Q \geq Q^{\text{pay}}$, we have $V(Q) \leq V(Q^{\text{pay}})$ for all $Q \geq Q^{\text{pay}}$. Suppose, toward a contradiction, that there is an adoption equilibrium $Q^* \geq Q^{\text{pay}}$. Then

$$Q^* = MF(V(Q^*)).$$

Because F is nondecreasing and $V(Q^*) \leq V(Q^{\text{pay}})$,

$$Q^* = MF(V(Q^*)) \leq MF(V(Q^{\text{pay}})).$$

The hypothesis

$$V(Q^{\text{pay}}) < F^{-1}\left(\frac{Q^{\text{pay}}}{M}\right)$$

implies

$$MF(V(Q^{\text{pay}})) < Q^{\text{pay}}.$$

Therefore $Q^* < Q^{\text{pay}}$, contradicting $Q^* \geq Q^{\text{pay}}$. No adoption equilibrium can support scale $Q \geq Q^{\text{pay}}$. $\square$

Derivative of adoption value. For the constant anticipated exit case in the main text,

$$V(Q) = m(Q) + (1 - \bar{f})y((1 - \bar{f})Q) - \bar{f}k(\bar{f}Q) - (1 - \tau)\frac{C(\bar{f}Q)}{Q}.$$

Differentiating gives

$$V'(Q) = m'(Q) + (1 - \bar{f})^2 y'((1 - \bar{f})Q) - \bar{f}^2 k'(\bar{f}Q) - (1 - \tau) \frac{\bar{f}Q C'(\bar{f}Q) - C(\bar{f}Q)}{Q^2}.$$

If C is convex and $C(0) = 0$, then $C(x) \leq xC'(x)$ for $x \geq 0$, so the adjustment-cost term is weakly negative. In the crowded-trade regime, $y' < 0$, so yield compression also reduces adoption value as scale rises.

General anticipated exit. If the anticipated exit fraction depends on scale, replace $\bar{f}$ by $\bar{f}(Q)$ and define $g(Q) = \bar{f}(Q)Q$ and $h(Q) = (1 - \bar{f}(Q))Q$. Then

$$V(Q) = m(Q) + (1 - \bar{f}(Q))y(h(Q)) - \bar{f}(Q)k(g(Q)) - (1 - \tau) \frac{C(g(Q))}{Q}.$$

The derivative is

$$\begin{aligned} V'(Q) = & m'(Q) - \bar{f}'(Q)y(h(Q)) + (1 - \bar{f}(Q))y'(h(Q))h'(Q) \\ & - \bar{f}'(Q)k(g(Q)) - \bar{f}(Q)k'(g(Q))g'(Q) \\ & - (1 - \tau) \frac{C'(g(Q))g'(Q)Q - C(g(Q))}{Q^2}, \end{aligned}$$

where $g'(Q) = \bar{f}(Q) + Q\bar{f}'(Q)$ and $h'(Q) = 1 - \bar{f}(Q) - Q\bar{f}'(Q)$. This more general expression has the same economic interpretation as the constant-exit formula, but the main text uses the constant-exit case to keep the extension transparent.